
Chapter 4: The Processor

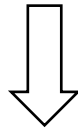
Ngo Lam Trung

[with materials from *Computer Organization and Design, 4th Edition*,
Patterson & Hennessy, © 2008, MK
and M.J. Irwin's presentation, PSU 2008]

Review

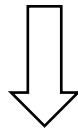
High-level
language
program
(in C)

```
swap(int v[], int k)
{int temp;
  temp = v[k];
  v[k] = v[k+1];
  v[k+1] = temp;
}
```



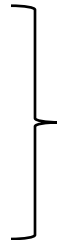
Assembly
language
program
(for MIPS)

```
swap:
    muli $2, $5, 4
    add  $2, $4, $2
    lw   $15, 0($2)
    lw   $16, 4($2)
    sw   $16, 0($2)
    sw   $15, 4($2)
    jr   $31
```



Binary machine
language
program
(for MIPS)

```
000000001010000100000000000011000
00000000000110000001100000100001
100011000110001000000000000000000
100011001111001000000000000000100
101011001111001000000000000000000
101011000110001000000000000000100
00000011111000000000000000001000
```



Performance metric

$$\text{CPU time} = \text{CPI} * \text{CC} * \text{IC}$$

CPI: cycle per instruction

CC: clock cycle

IC: instruction count

How to improve?

- IC:
- CC:
- CPI:

In this chapter

- Implementation of data path
- How to get $\text{CPI} < 1$

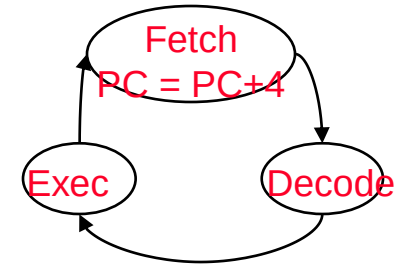
Overview

- ❑ We will examine two MIPS implementations
 - ❑ A simplified version
 - ❑ A more realistic pipelined version
- ❑ Limit to a simple subset of MIPS ISA
 - ❑ Memory reference: lw, sw
 - ❑ Arithmetic/logical: add, sub, and, or, slt
 - ❑ Control transfer: beq, j
- ❑ Implementation of real CPU with other instructions are similar to the simplified version (theoretically!)

General instruction cycle

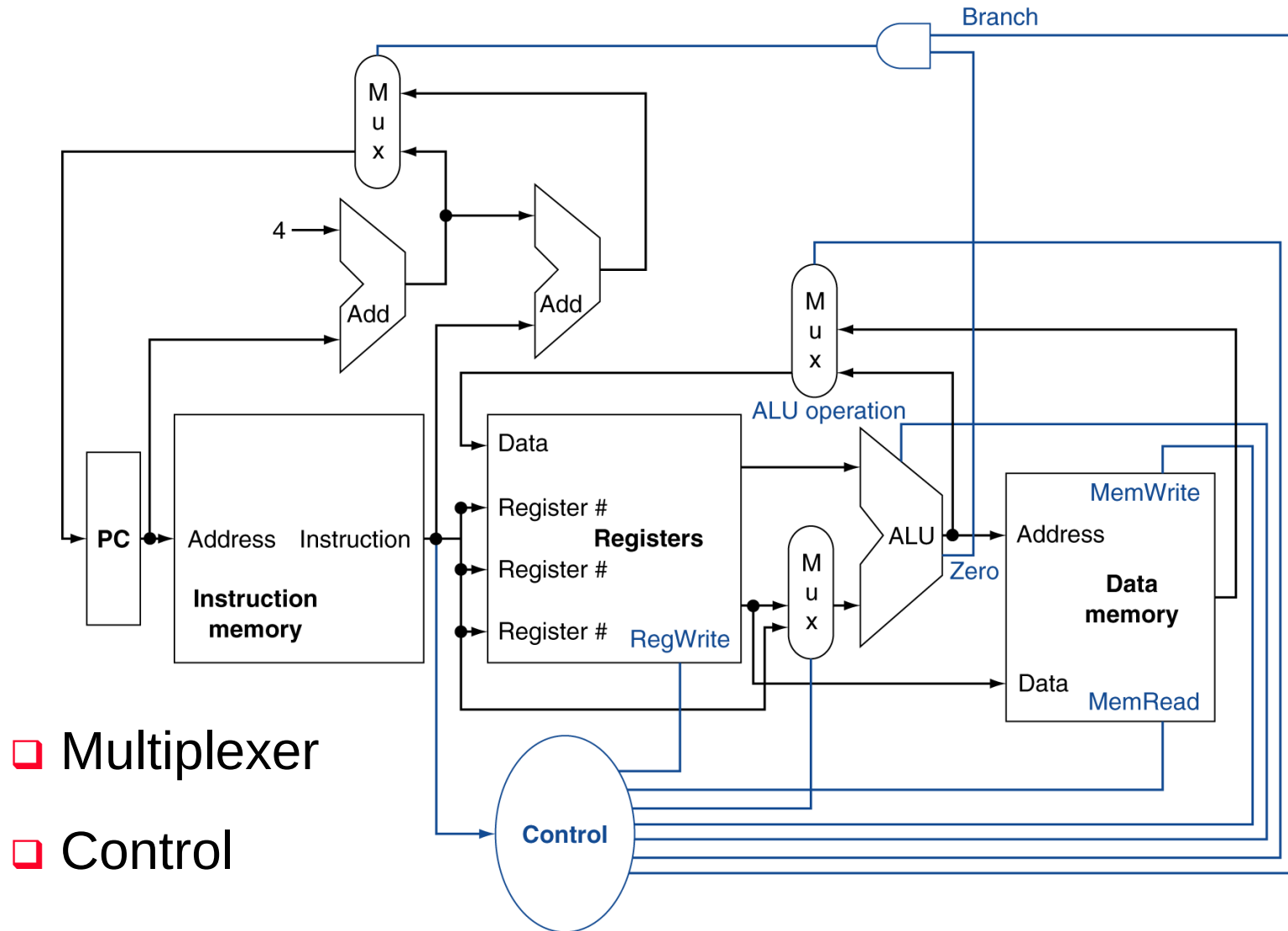
❑ Generic implementation

- ❑ use the program counter (PC) to supply the instruction address and fetch the instruction from memory (and update the PC)
- ❑ decode the instruction (and read registers)
- ❑ execute the instruction



- ❑ All instructions (except **j**) use the ALU after reading the registers
 - ❑ ALU: Arithmetic and Logic Unit, where the arithmetic and logic operations are executed
- ❑ In this chapter: implementation of CPU that can execute the simple subset of MIPS ISA

CPU implementation with MUXes and Control



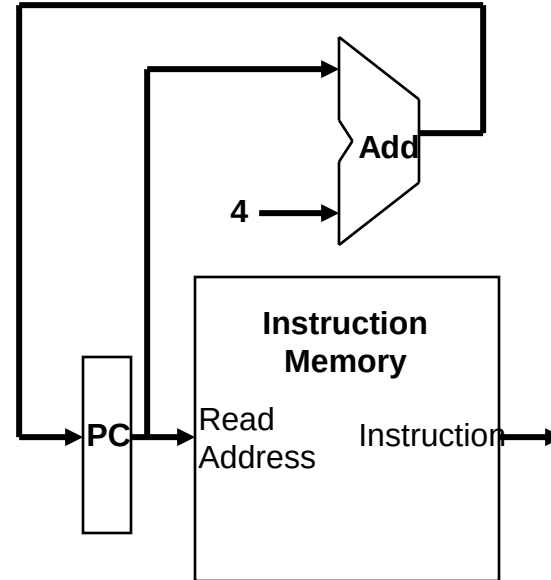
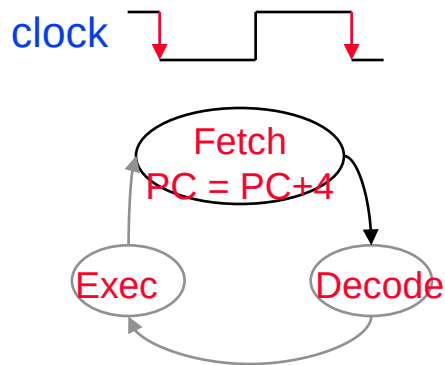
❑ Multiplexer

❑ Control

Don't panic! We'll build this incrementally.

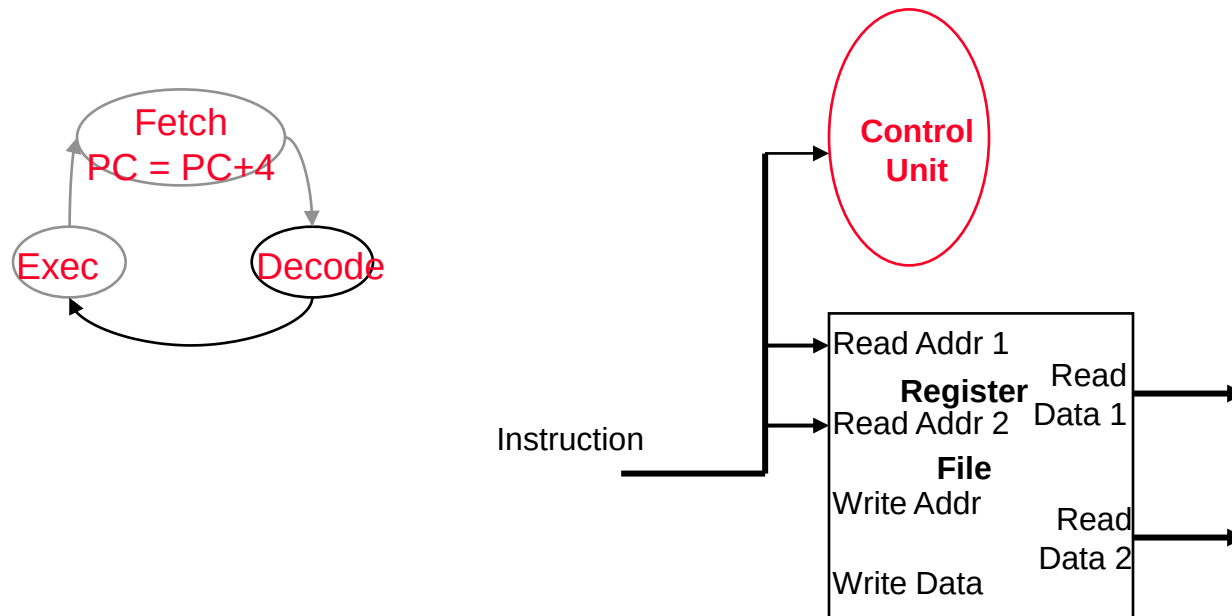
Fetching Instructions

- ❑ Fetching instructions involves
 - ❑ reading the instruction from the Instruction Memory
 - ❑ updating the PC value to be the address of the next instruction in memory



Decoding Instructions

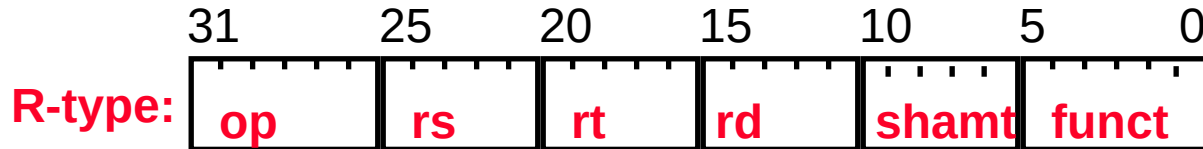
- ❑ Decoding instructions involves
 - ❑ sending the fetched instruction's opcode and function field bits to the control unit
 - ❑ The control unit send appropriate control signals to other parts inside CPU to execute the operations corresponds to the instruction



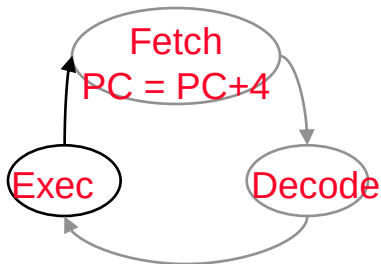
- Example: reading two values from the Register File
→ Register File addresses are contained in the instruction

Executing R Format Operations

□ R format operations (**add**, **sub**, **slt**, **and**, **or**)



- read two register operands **rs** and **rt**
- perform operation (**op** and **funct**) on values in **rs** and **rt**
- store the result back into the Register File (into location **rd**)

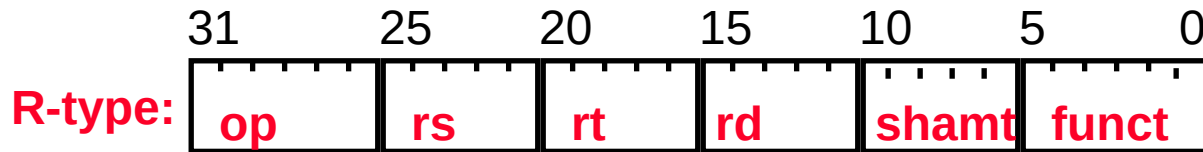


Example: **add s1, s2, s3**

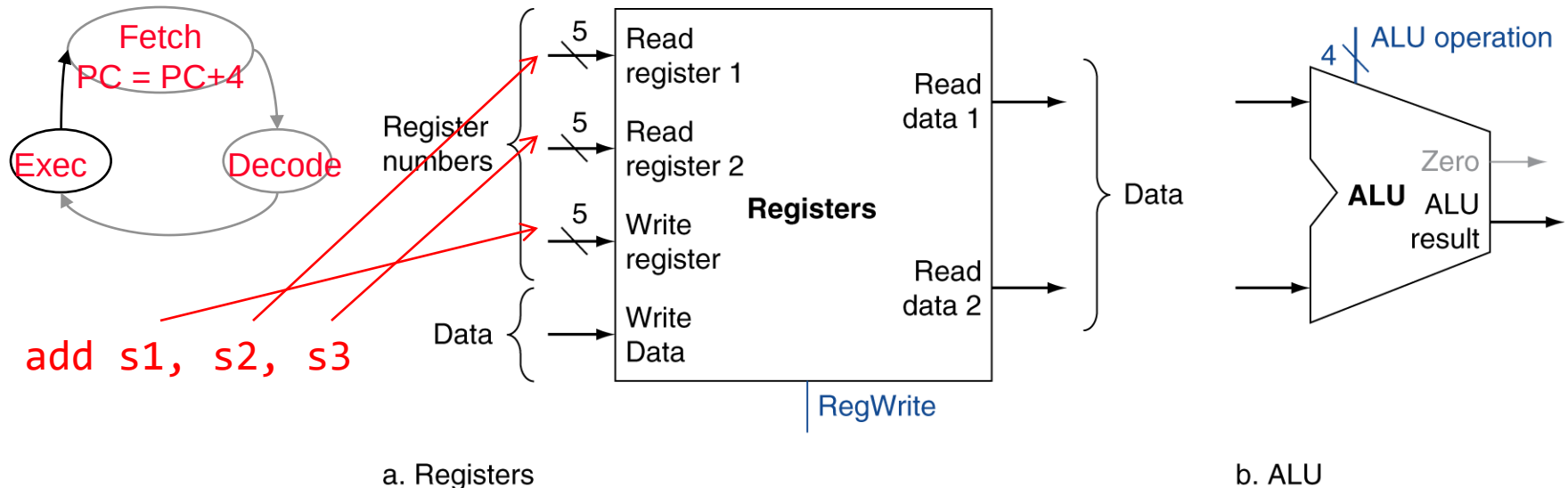
- Value of s2 and s3 are sent to ALU
- ALU execute the s2 + s3 operation
- Result is store into s1

Executing R Format Operations

❑ R format operations (**add**, **sub**, **slt**, **and**, **or**)



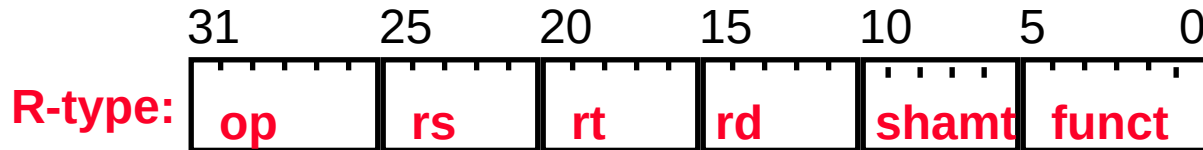
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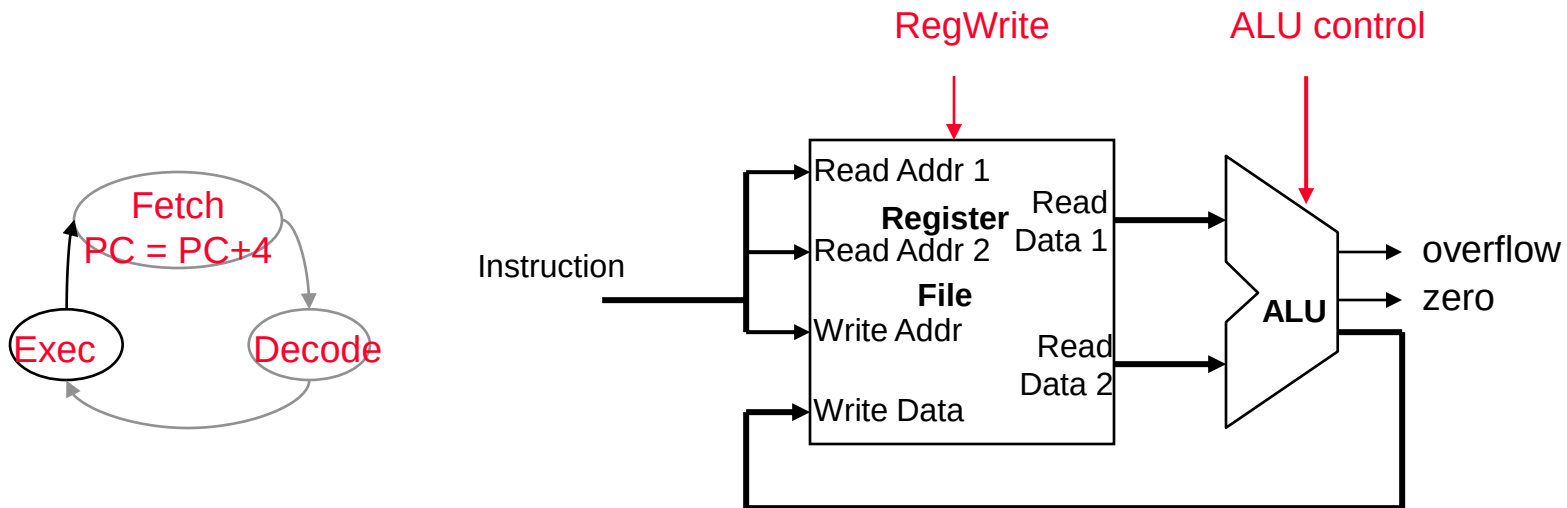
Draw connection between a and b to form the execution unit?

Executing R Format Operations

□ R format operations (**add**, **sub**, **slt**, **and**, **or**)



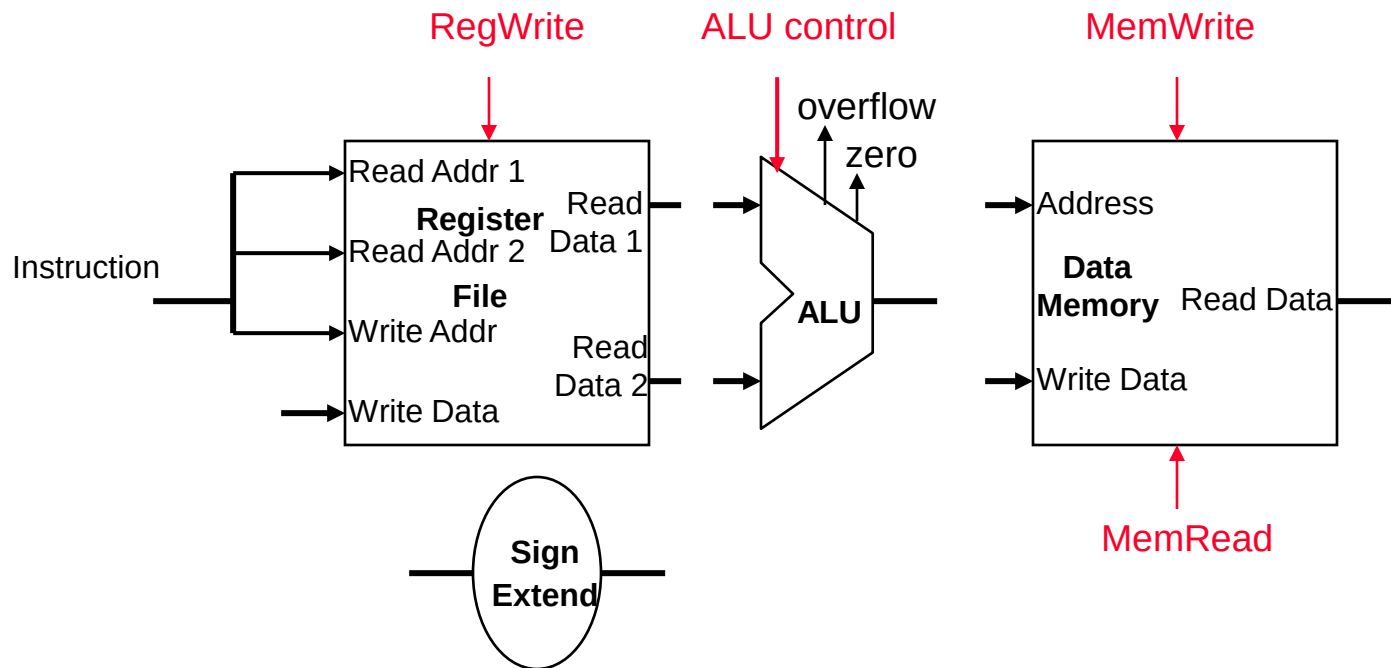
- read two register operands **rs** and **rt**
- perform operation (**op** and **funct**) on values in **rs** and **rt**
- store the result back into the Register File (into location **rd**)



- We need the **write control signal** to control when the result is written to Register File

Executing Load and Store Operations

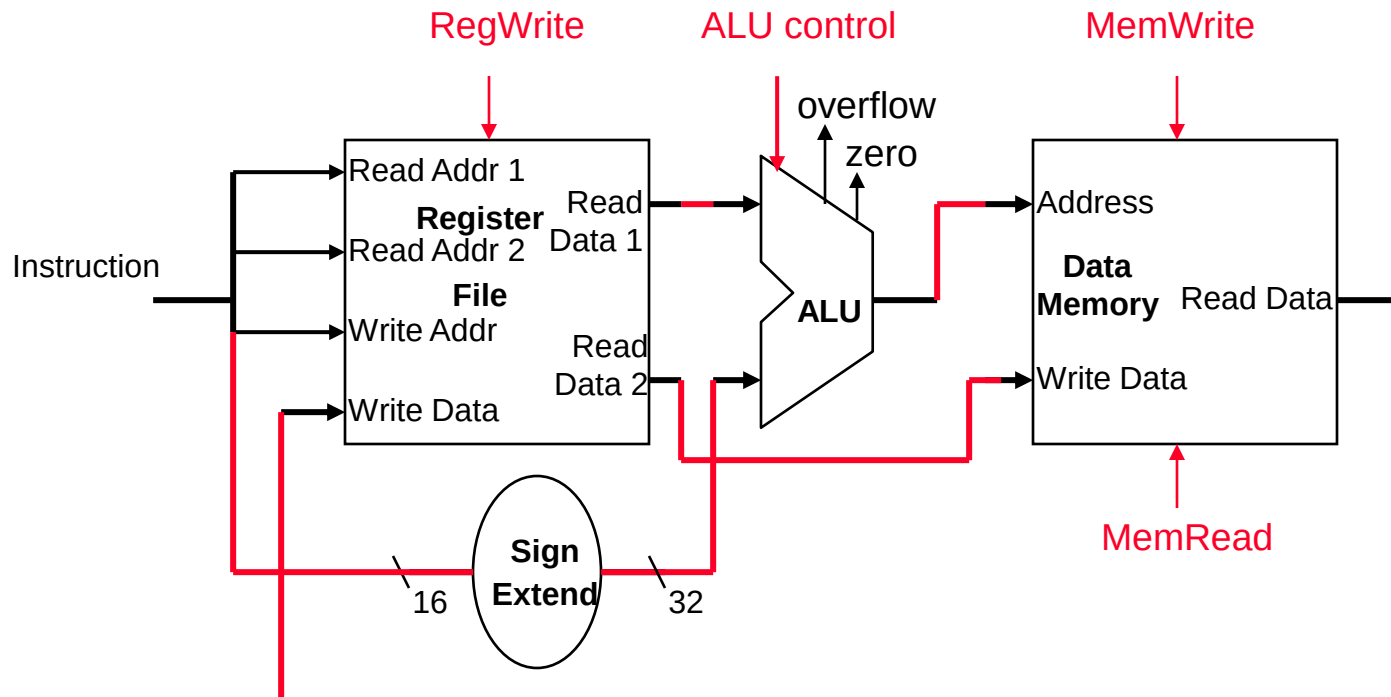
- ❑ Load and store operations involves
 - ❑ read register operands (including one base register)
 - ❑ compute memory address by adding the base to the offset
 - The 16-bit offset field in the instruction is signed-extended to 32 bit
 - ❑ **store**: read from the Register File, write to the Data Memory
 - ❑ **load**: read from the Data Memory, write to the Register File



Draw necessary connections to form execution unit?

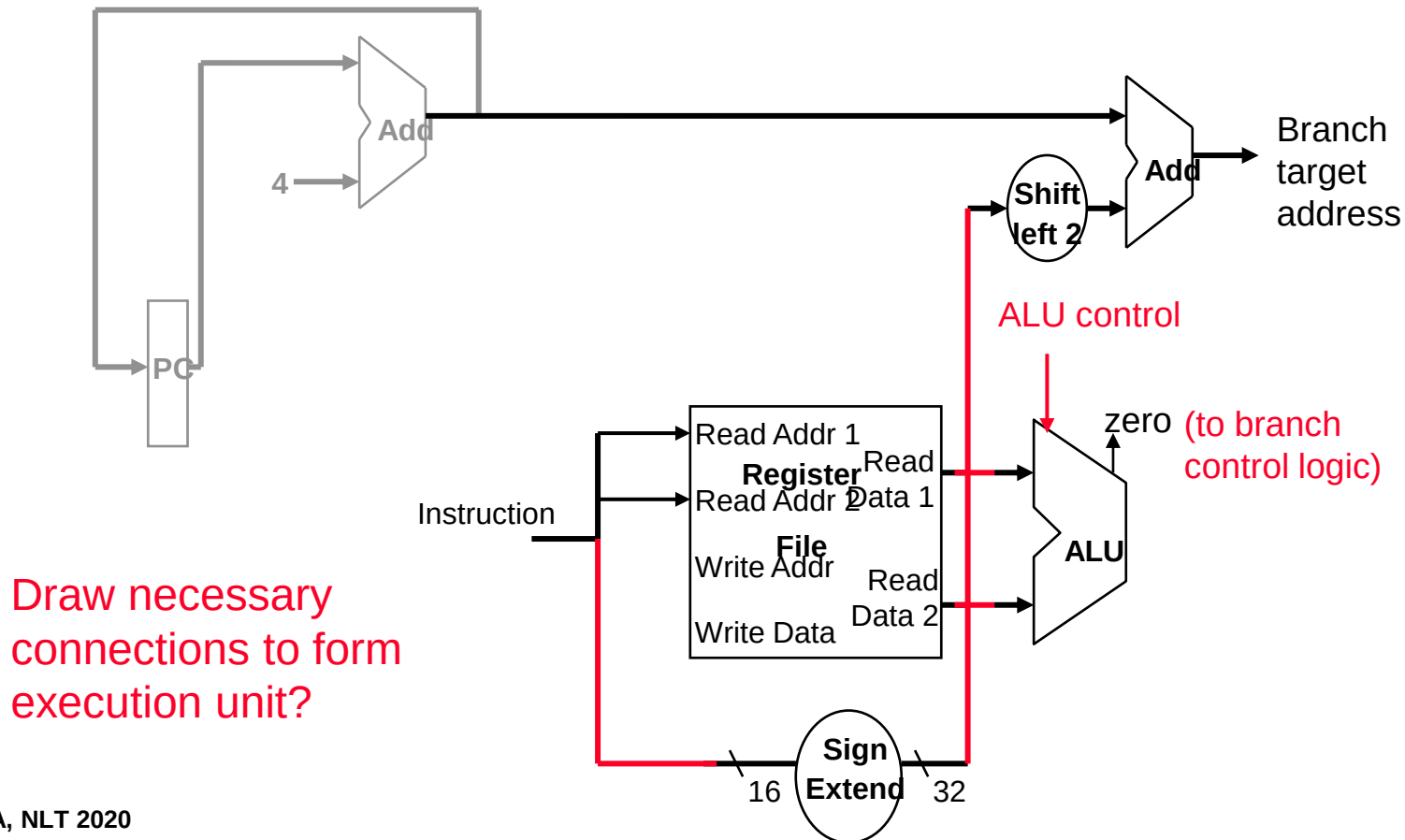
Executing Load and Store Operations

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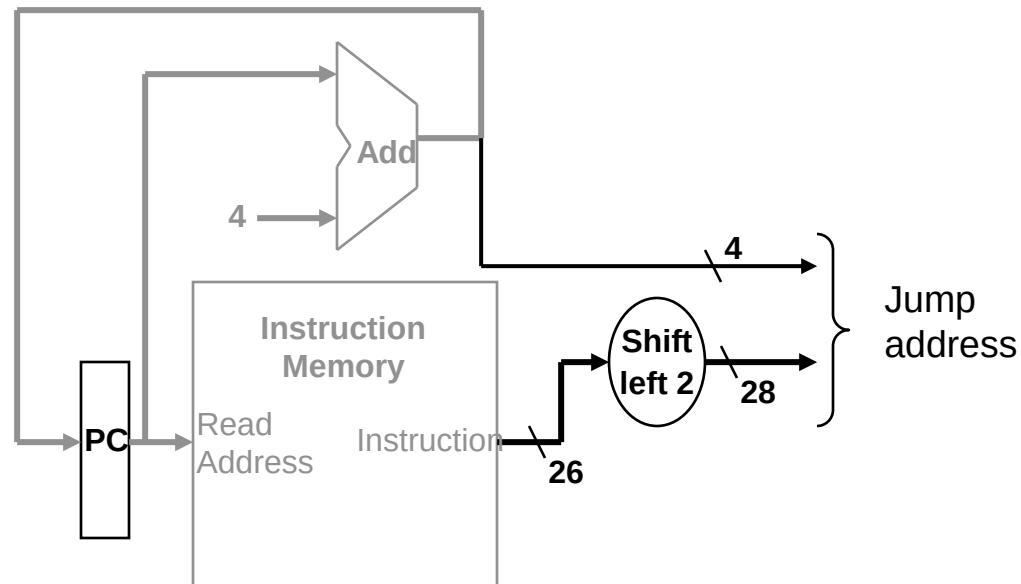
Executing Branch Operations

- ❑ Branch operations involves
 - ❑ read register operands
 - ❑ compare the operands (subtract, check **zero** ALU output)
 - ❑ compute the branch target address: adding the updated PC to the 16-bit signed-extended offset field in the instr



Executing Jump Operations

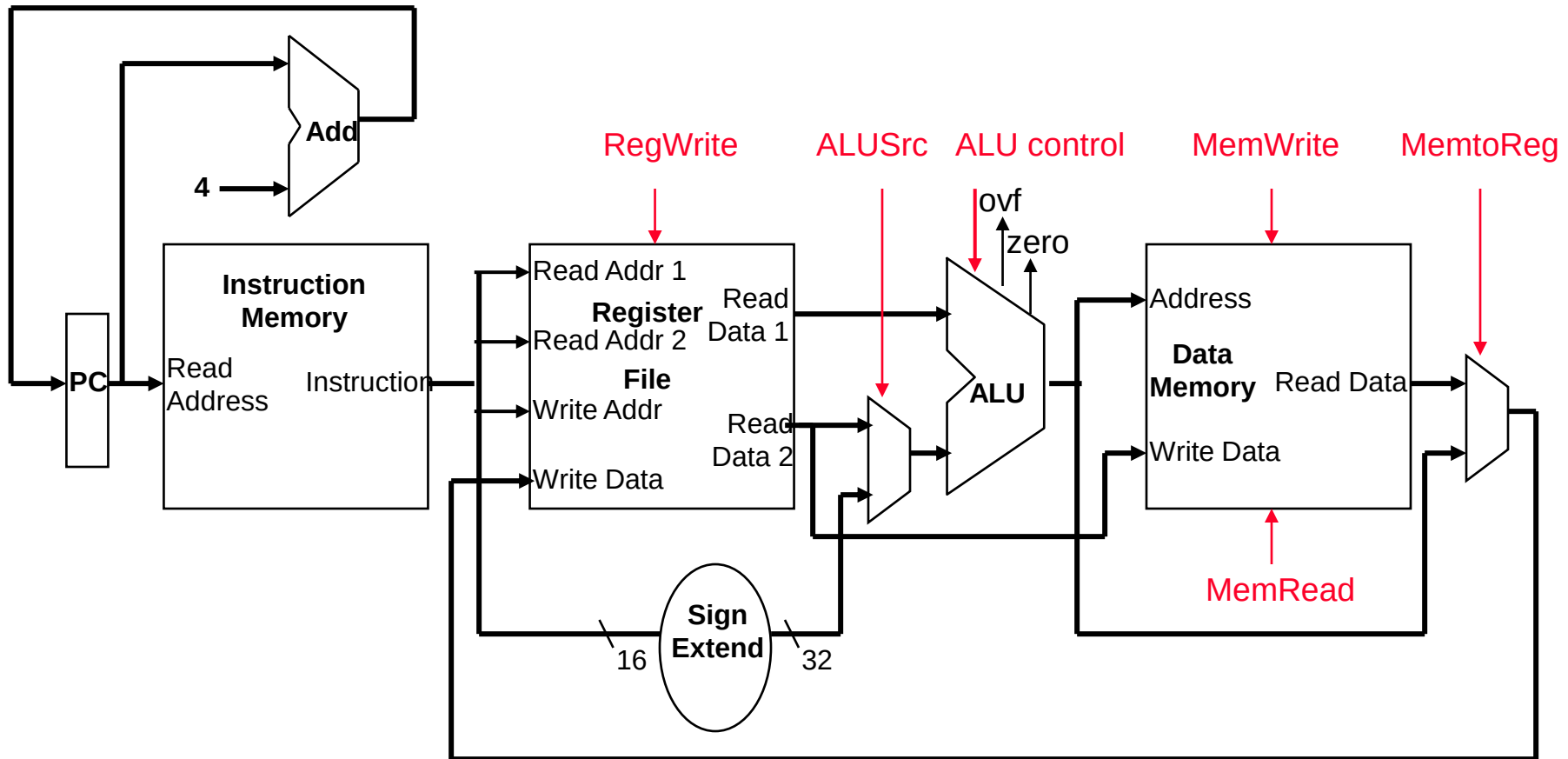
- ❑ Jump operation involves
 - ❑ keep 4 highest bits of PC
 - ❑ replace the lower 28 bits of the PC by
 - the lower 26 bits of the fetched instruction shifted left by 2 bits



Creating a Single Datapath from the Parts

- ❑ Assemble the datapath segments and add control lines and multiplexors as needed
- ❑ **Single cycle** design – fetch, decode and execute each instructions in **one** clock cycle
 - ❑ separate Instruction Memory and Data Memory, though they are both in main memory
 - ❑ **multiplexors** needed at the input of shared elements with control lines to do the selection
 - ❑ write signals to control writing to the Register File and Data Memory

Fetch, R, and Memory Access Portions



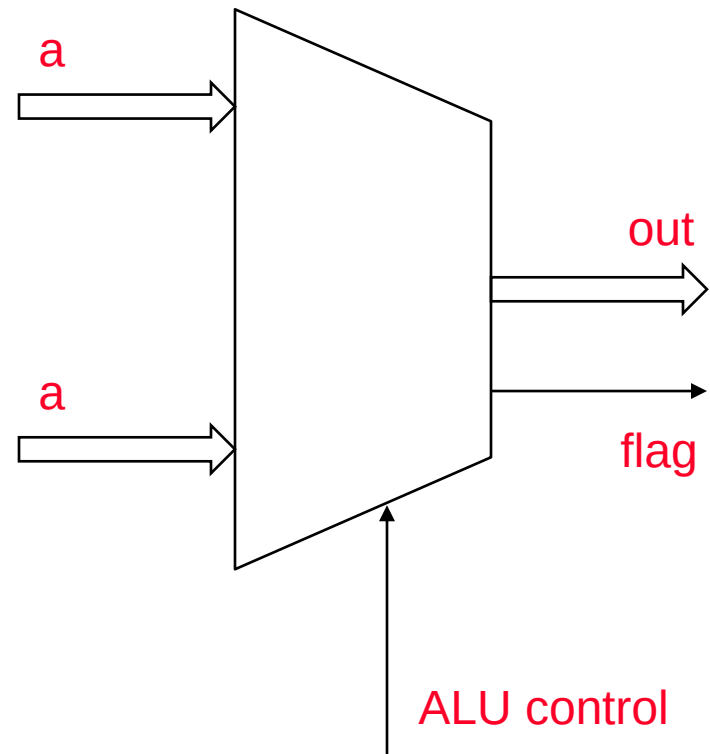
Designing ALU

❑ Input/output

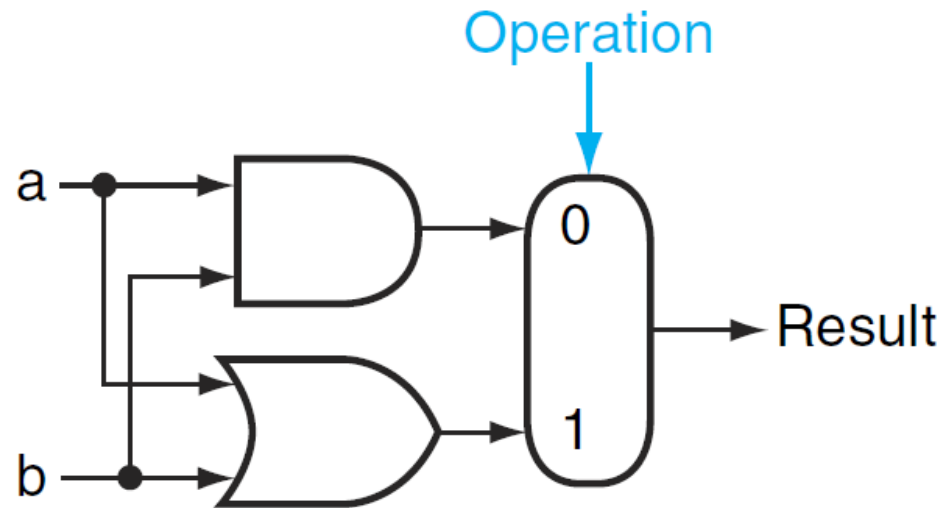
- ❑ Two data input: a, b
- ❑ ALU control signal
- ❑ Data out
- ❑ Flags out

❑ Operations

- ❑ And, or, nor
- ❑ Add, subtract



1-bit ALU with logic operation

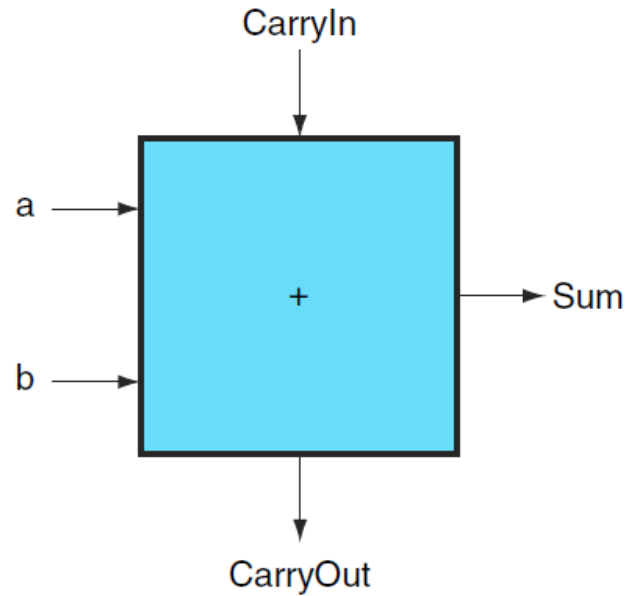


❑ What do we have if

❑ Operation = 0:

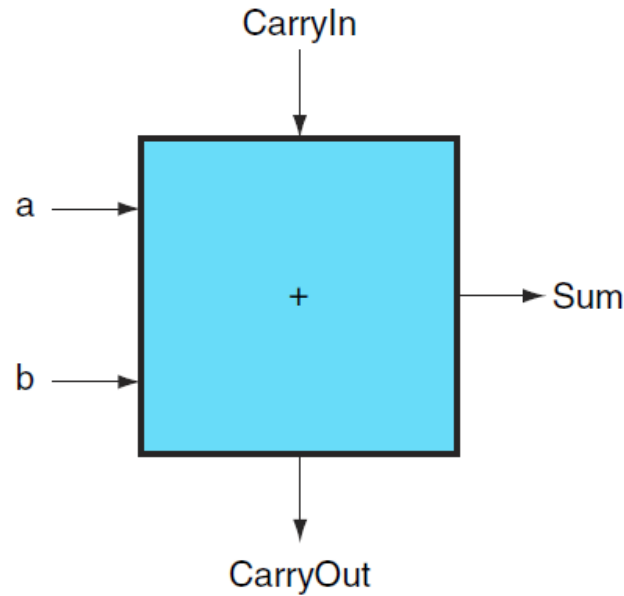
❑ Operation = 1:

1-bit full-adder



Inputs			Outputs	
a	b	CarryIn	CarryOut	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

1-bit full-adder

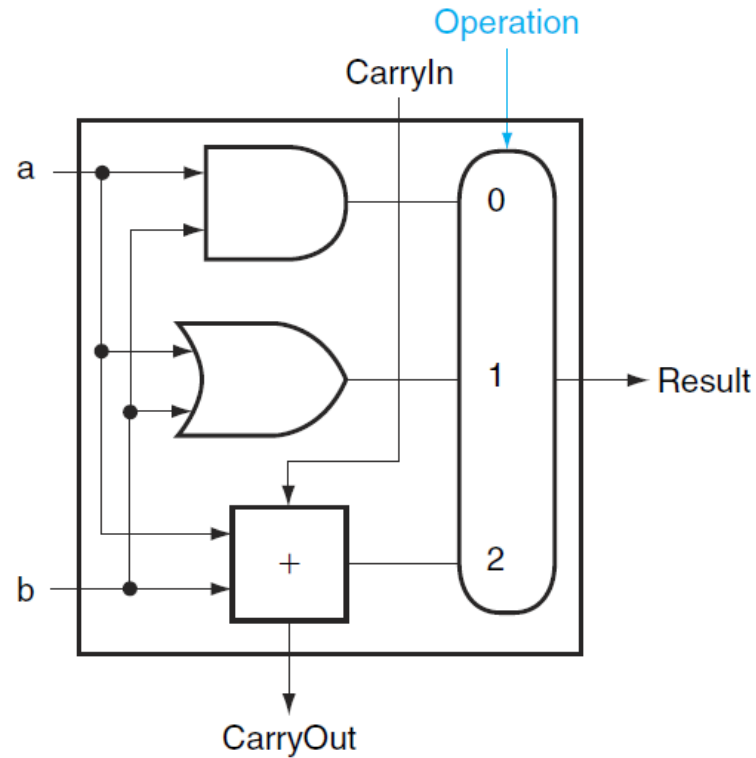


$$\text{CarryOut} = (b \cdot \text{CarryIn}) + (a \cdot \text{CarryIn}) + (a \cdot b)$$

$$\text{Sum} = (a \cdot \bar{b} \cdot \overline{\text{CarryIn}}) + (\bar{a} \cdot b \cdot \overline{\text{CarryIn}}) + (\bar{a} \cdot \bar{b} \cdot \text{CarryIn}) + (a \cdot b \cdot \text{CarryIn})$$

➔ Combinational circuit for full-adder can be constructed

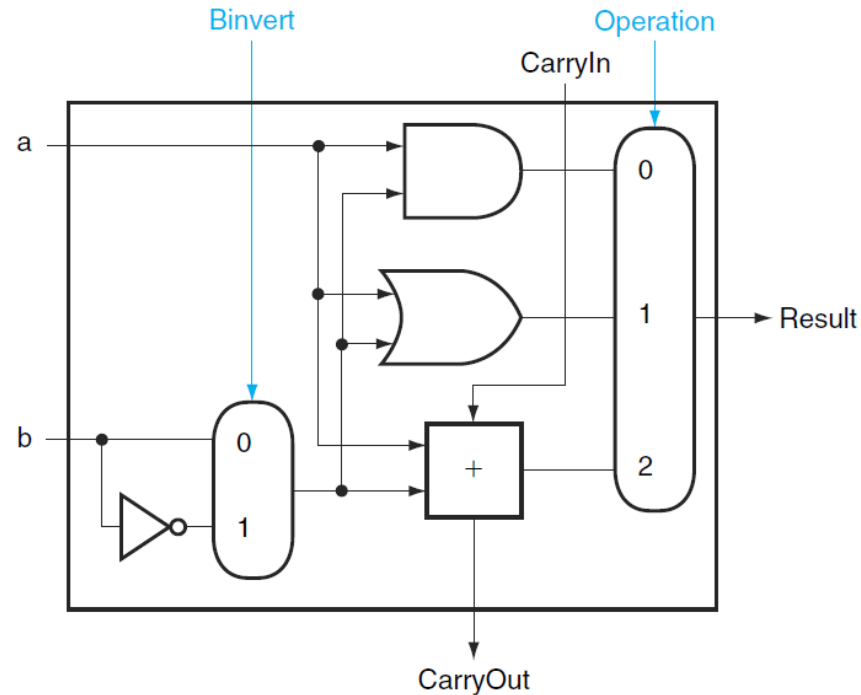
1-bit ALU with AND, OR, ADD



- ❑ Operation = 00:
- ❑ Operation = 01:
- ❑ Operation = 10:

How about 1-bit ALU with AND, OR, ADD, SUB?

□ $a - b = a + (-b) = a + (2\text{'s complement of } b)$



□ For SUB operation

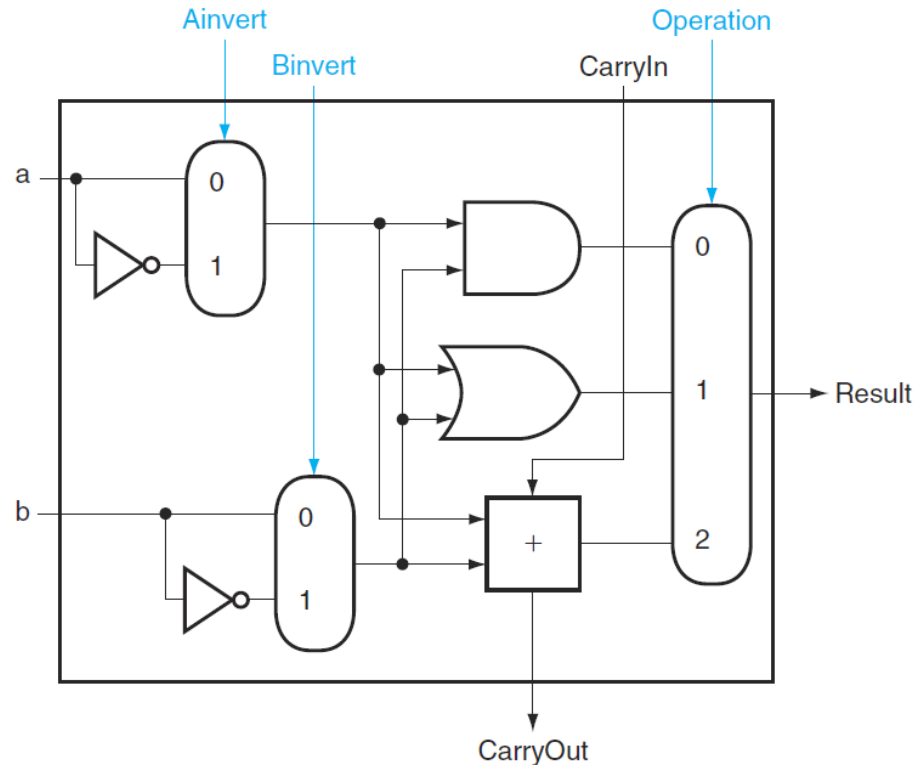
□ Operation =

□ Binvert =

□ CarryIn =

How to add NOR operation?

❑ $\overline{a + b} = \bar{a} \cdot \bar{b}$

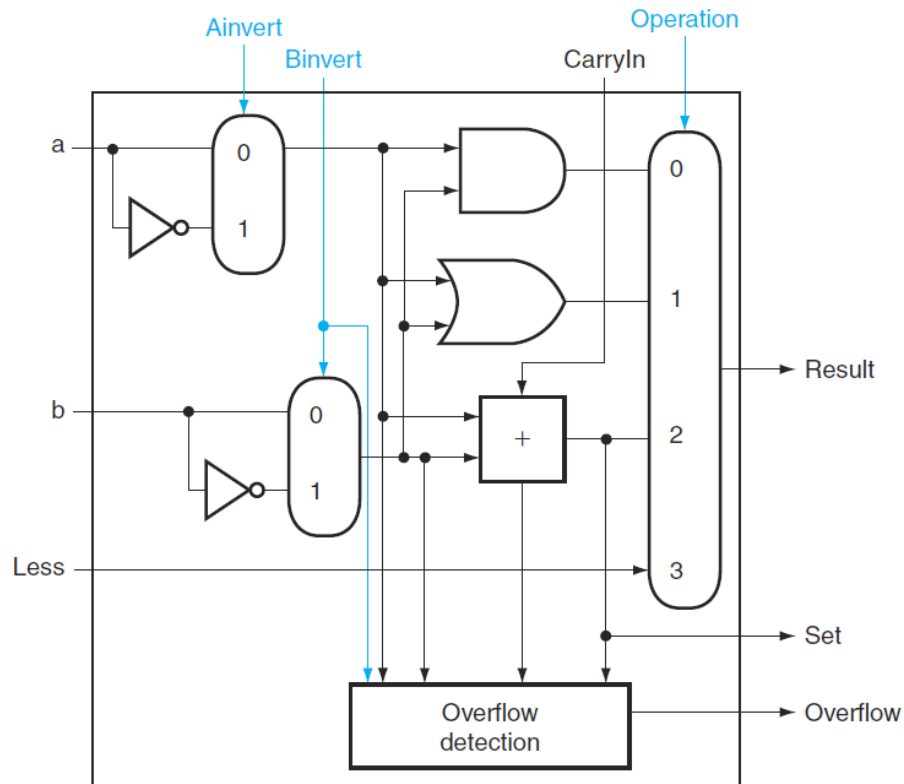


❑ Find control signal for NOR operation:

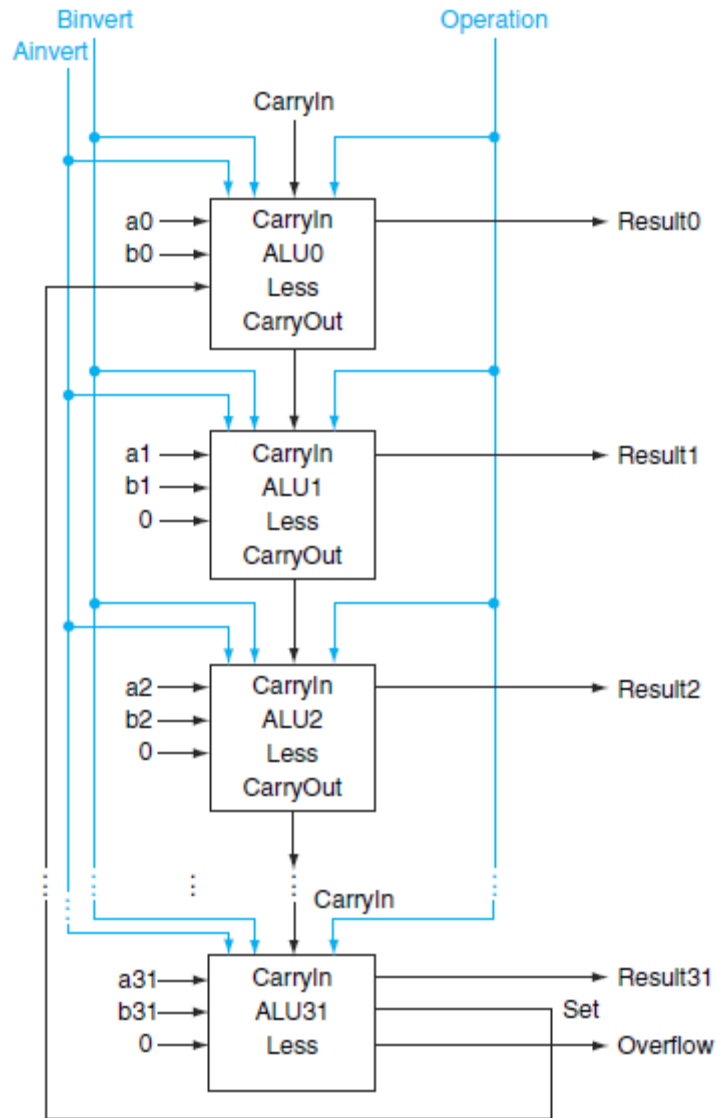
Adding SLT operation

❑ Check highest bit of (a-b)

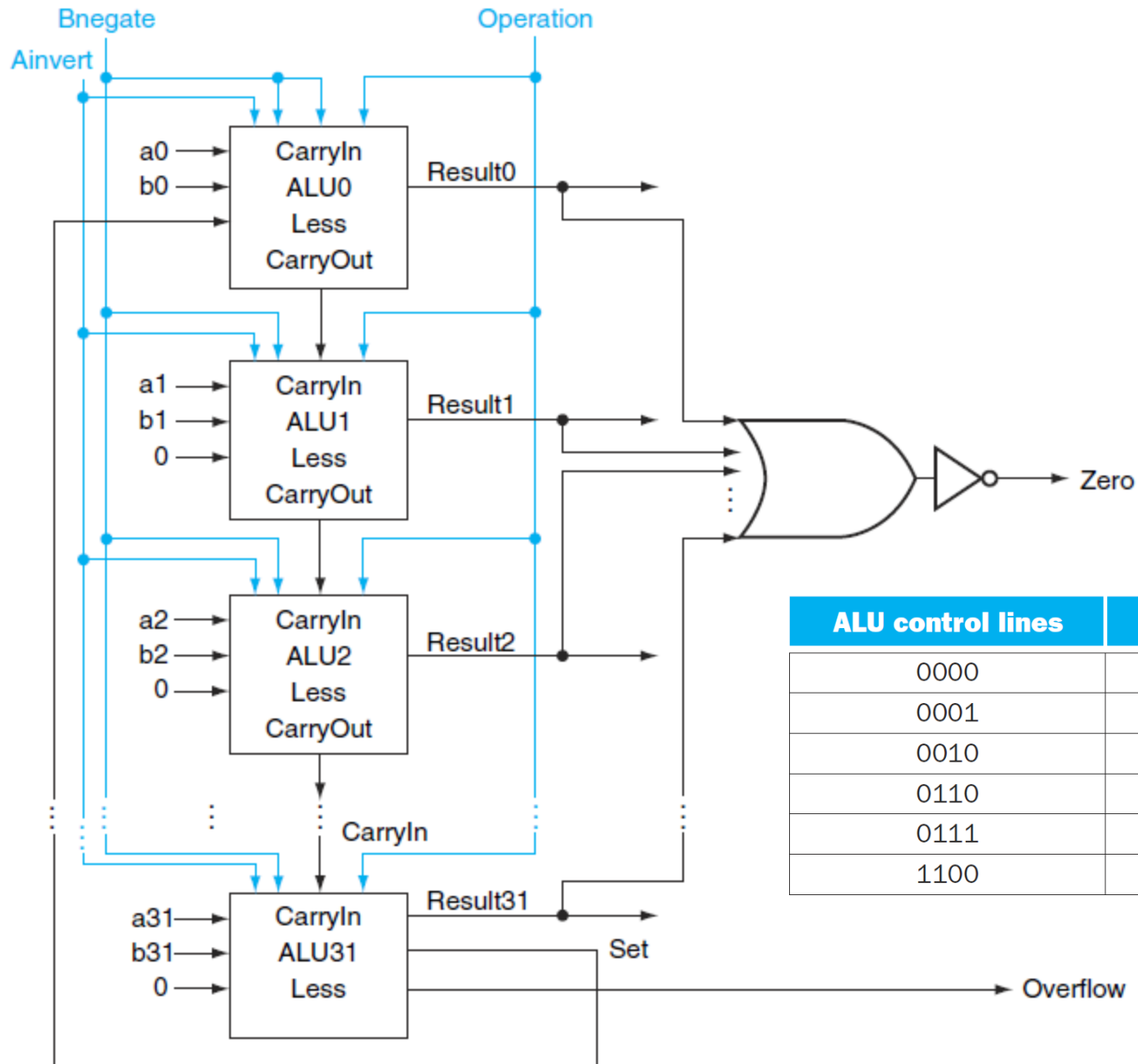
- ❑ 1: $a < b \rightarrow$ set
- ❑ 0: $a \geq b \rightarrow$ clear



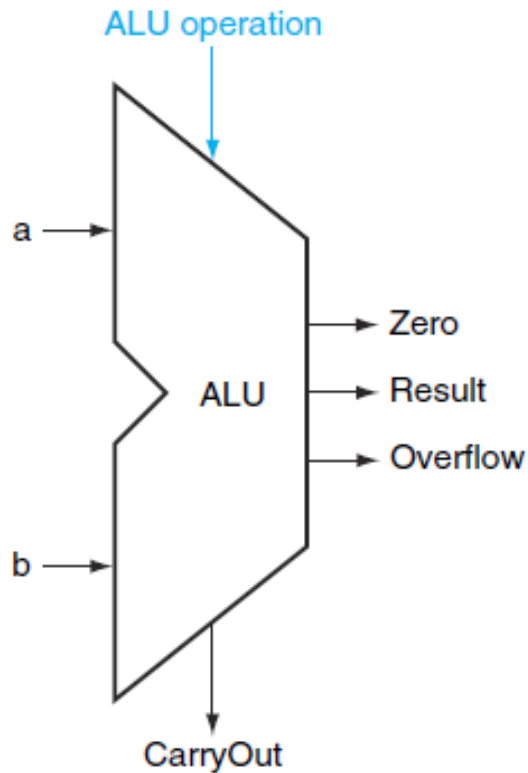
Building 32-bit ALU from 1-bit ALUs



Final ALU with AND, OR, NOR, ADD, SUB, SLT

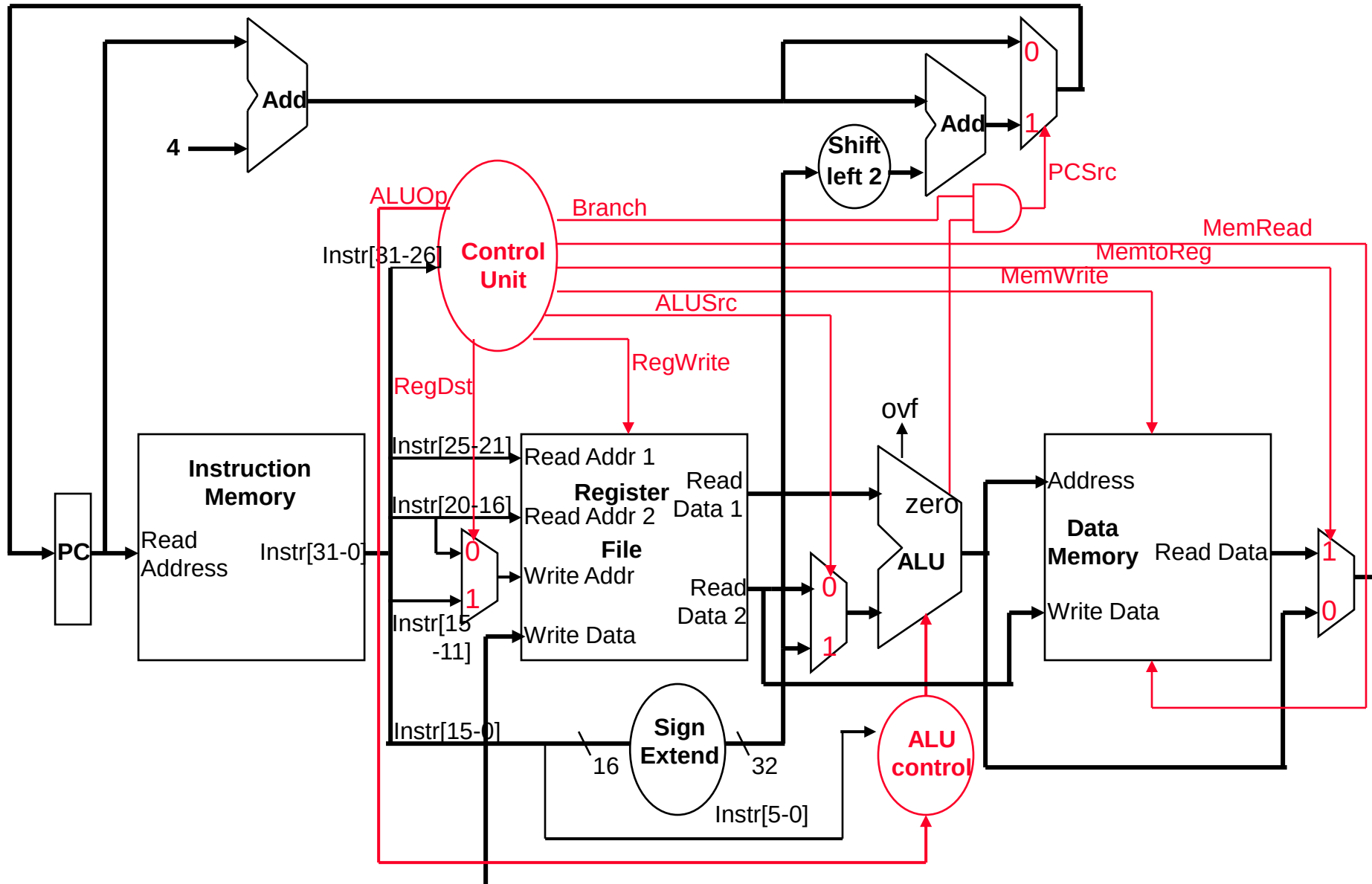


ALU symbol and interconnection

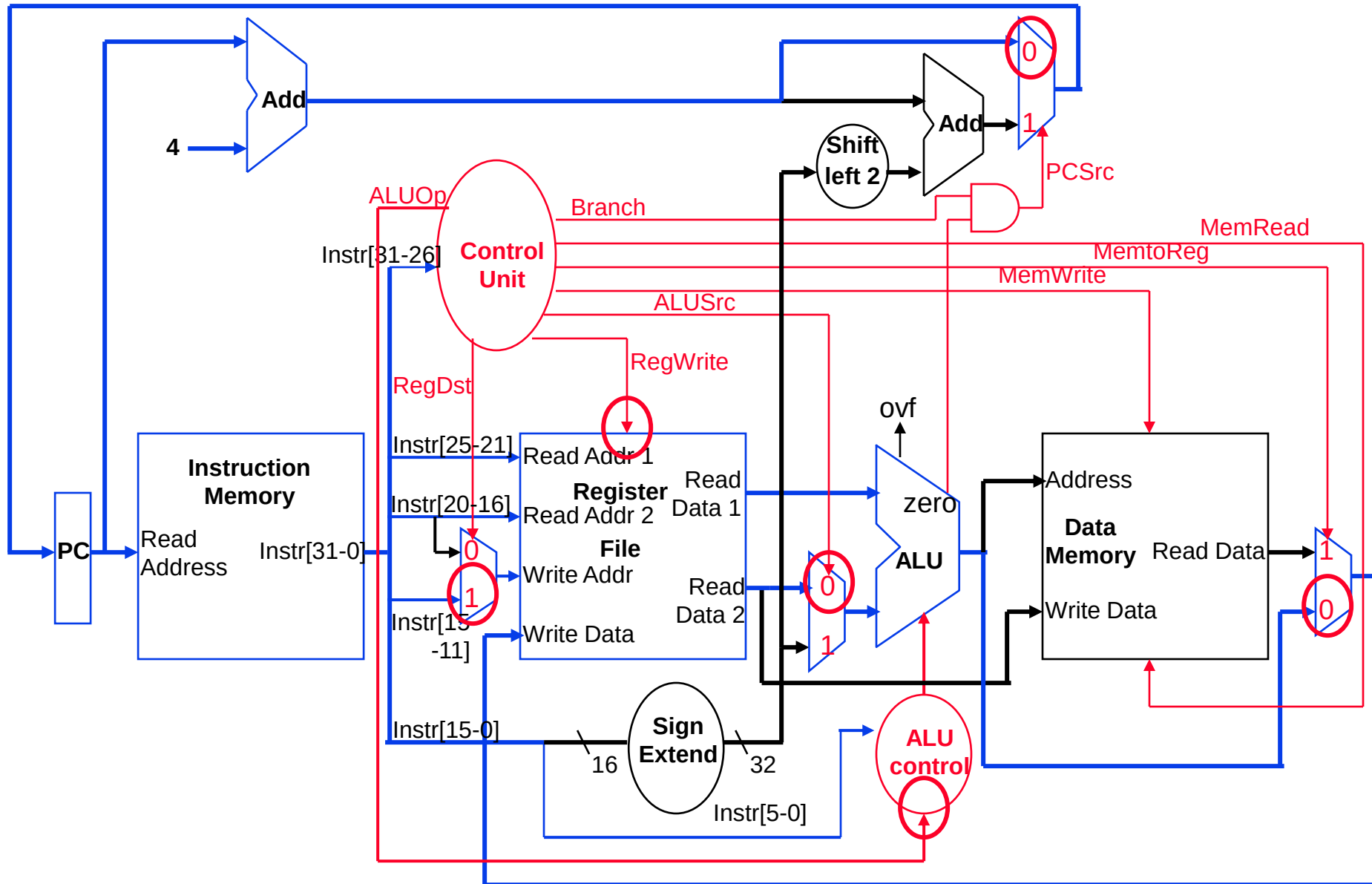


ALU control lines	Function
0000	AND
0001	OR
0010	add
0110	subtract
0111	set on less than
1100	NOR

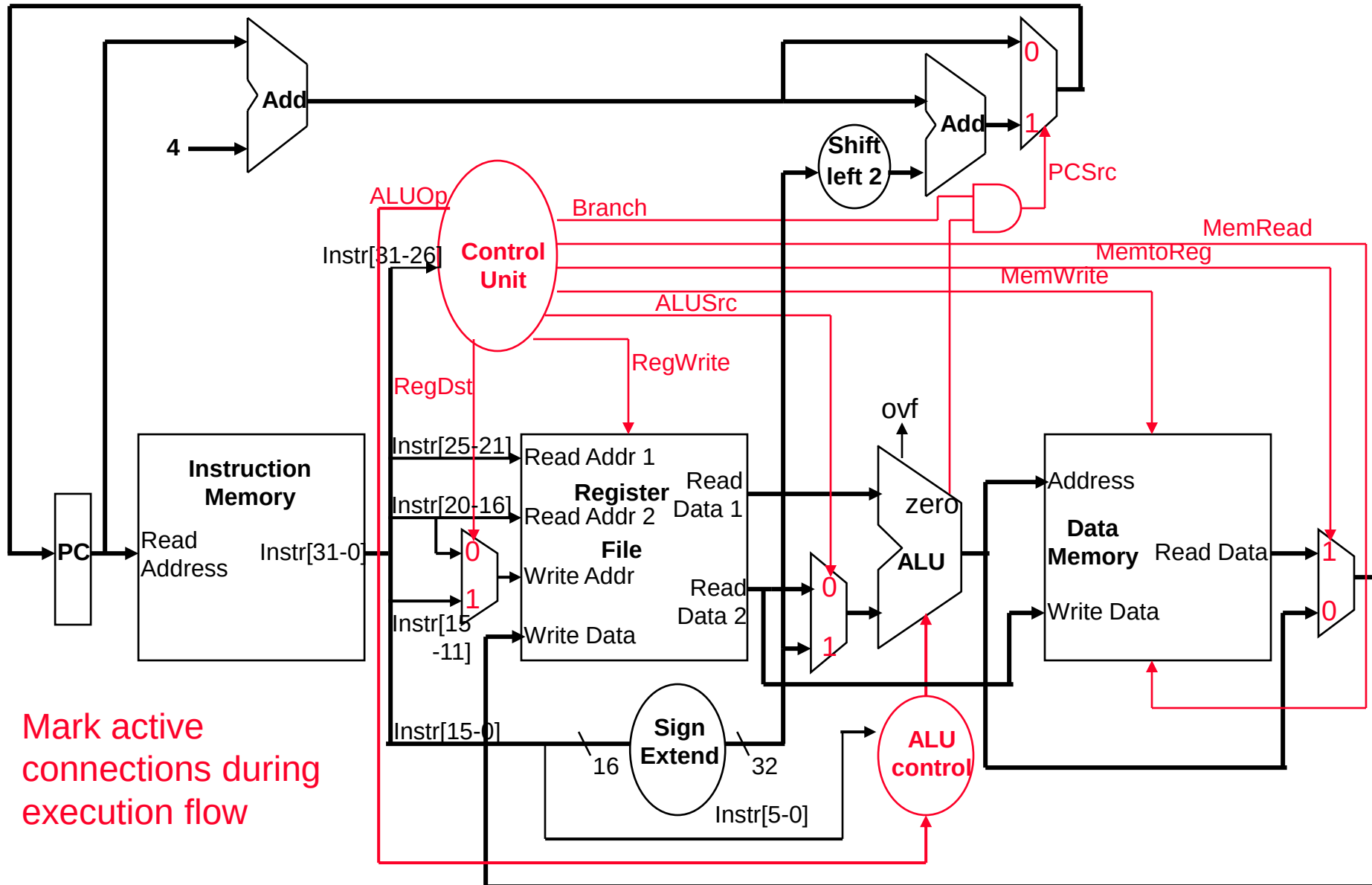
Adding Control Unit to build single datapath



R-type Instruction Data/Control Flow

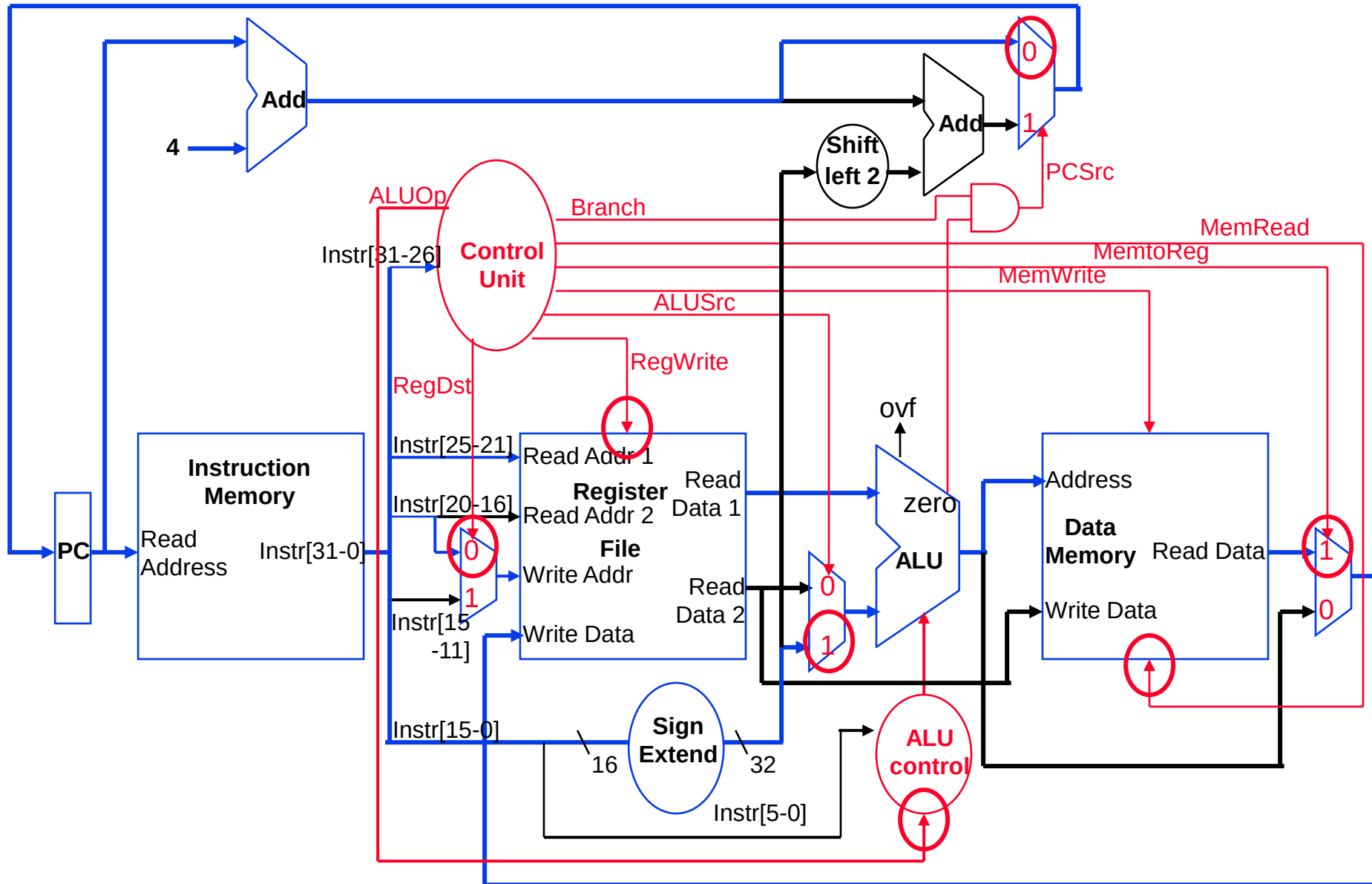


Load Word Instruction Data/Control Flow

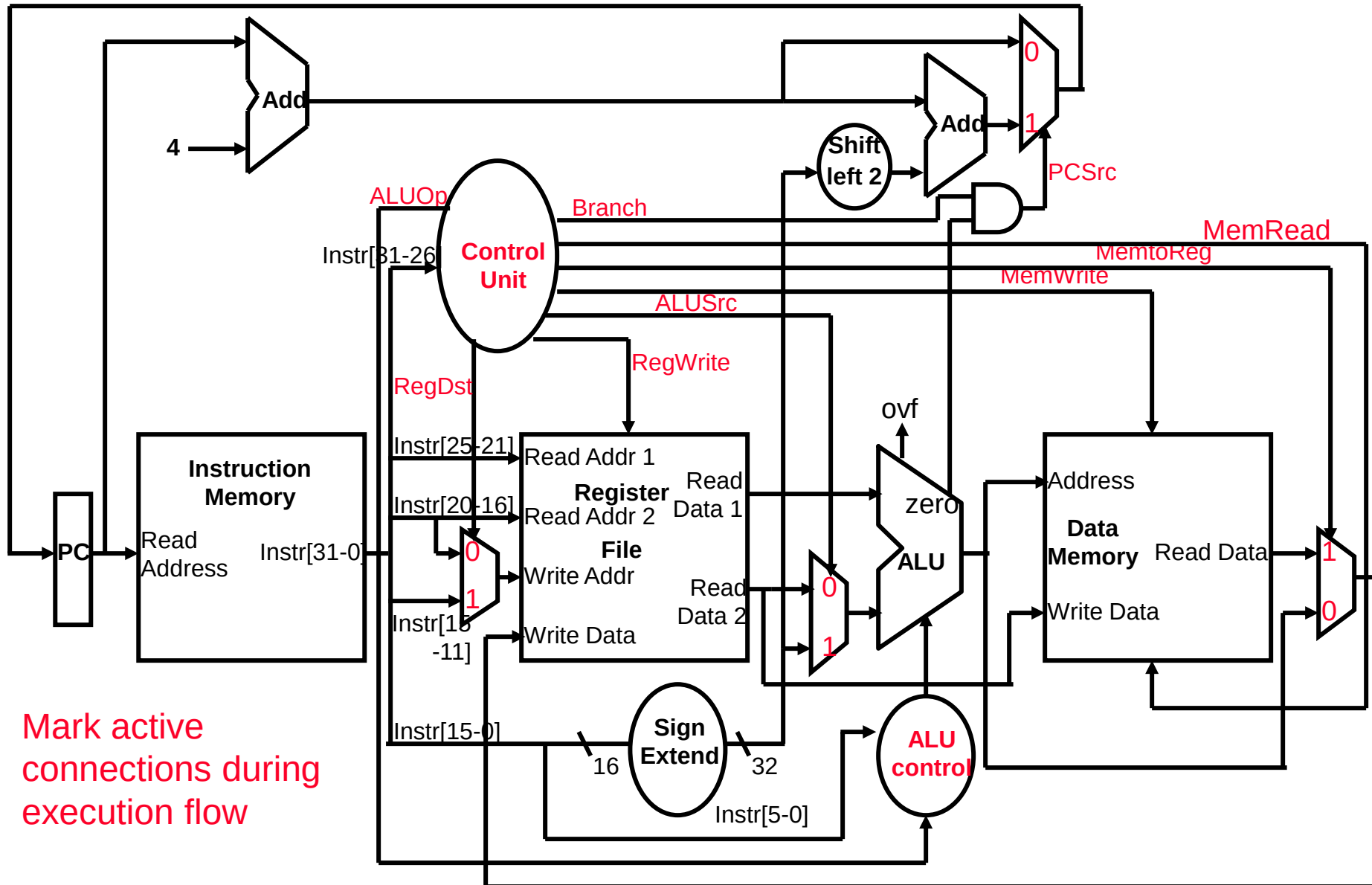


Mark active connections during execution flow

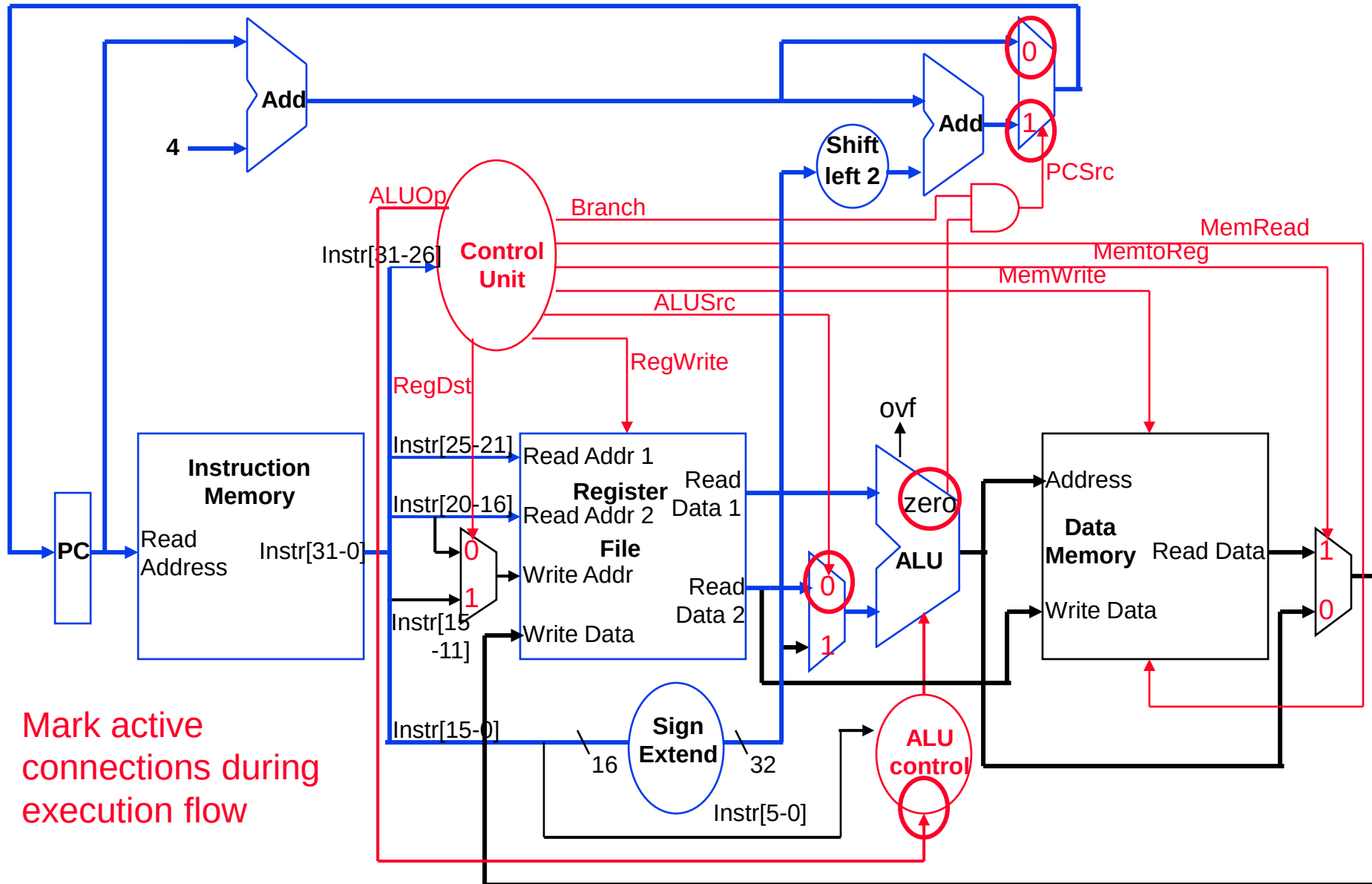
Load Word Instruction Data/Control Flow



Branch Instruction Data/Control Flow

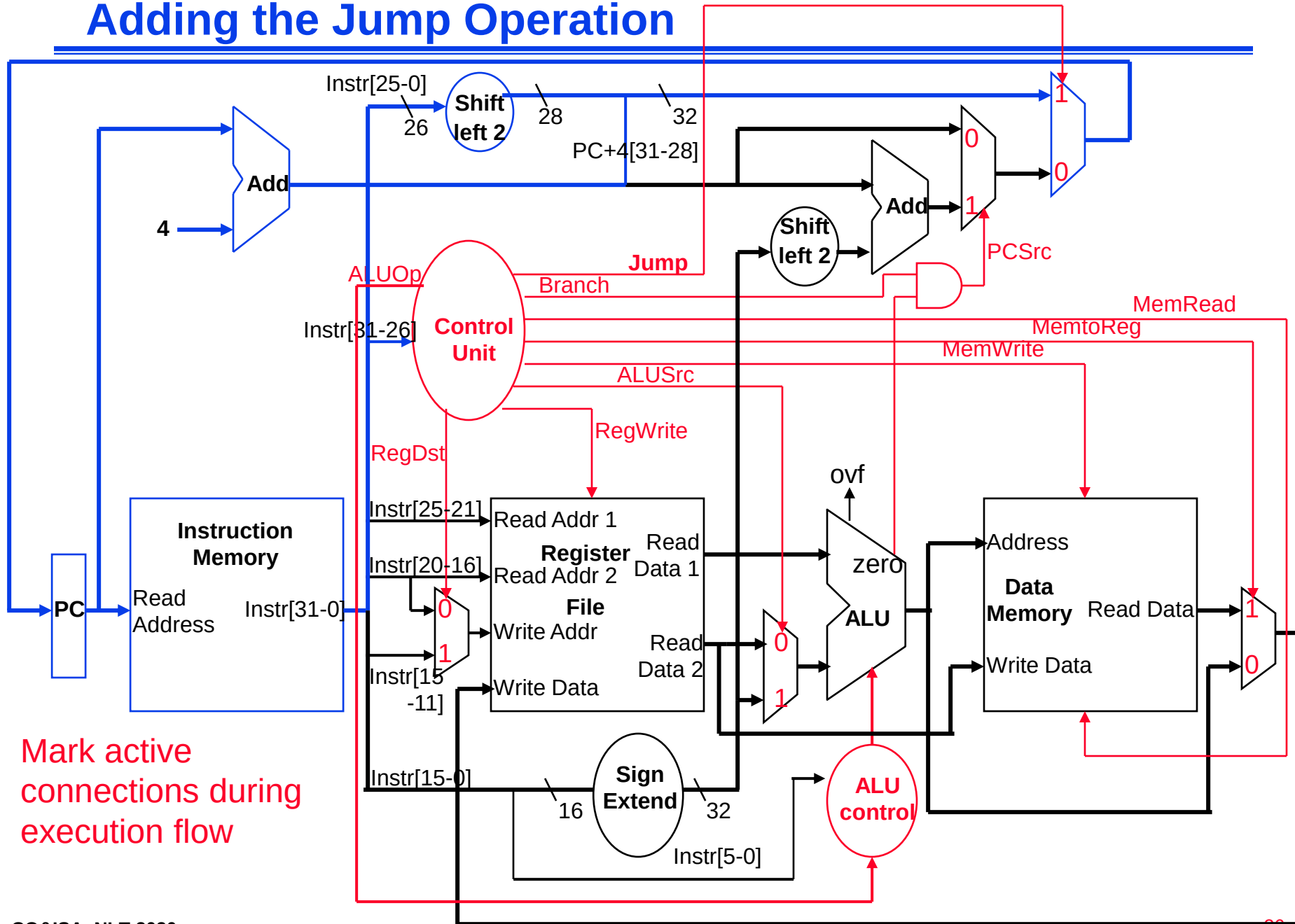


Branch Instruction Data/Control Flow

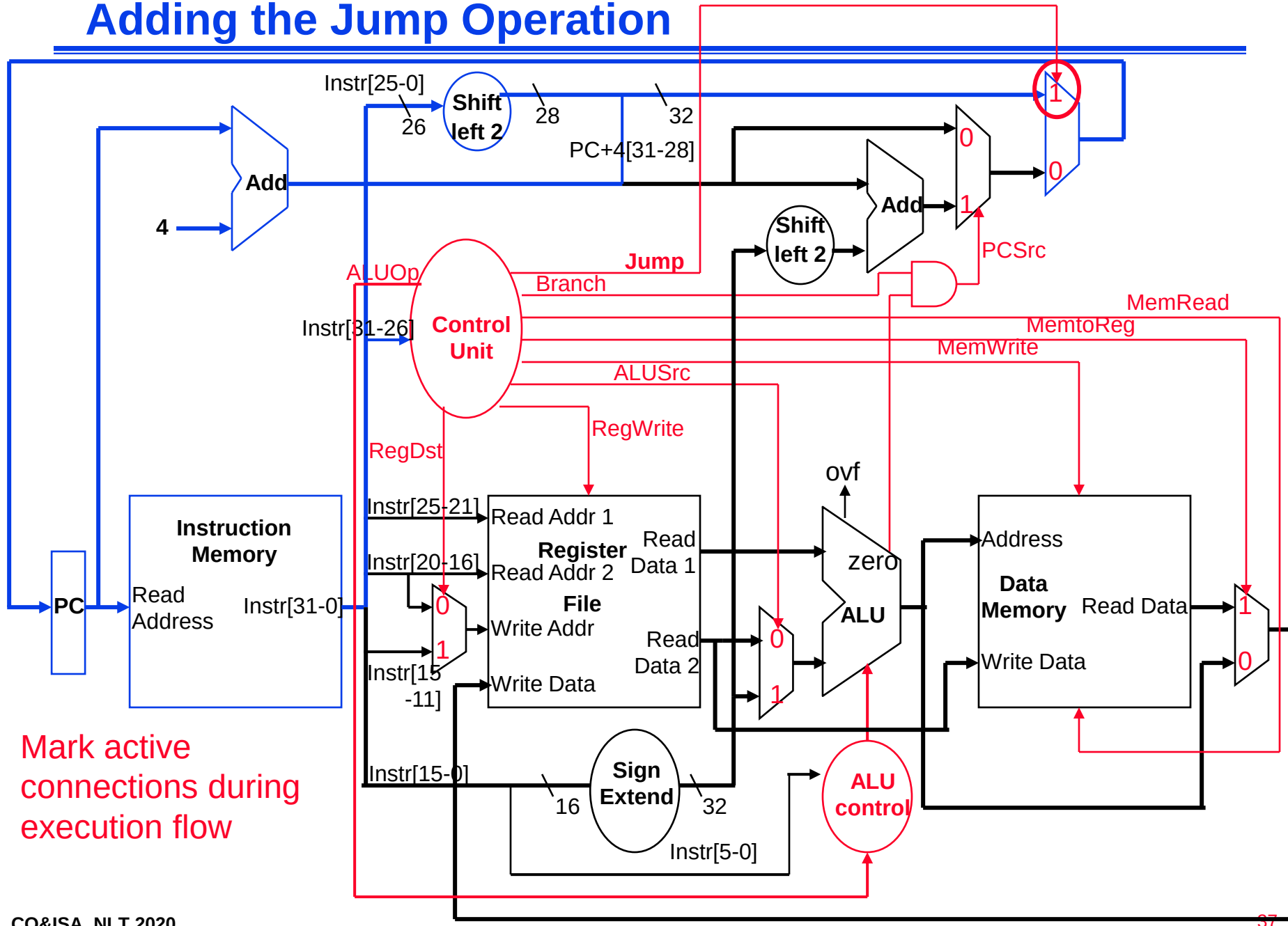


Mark active connections during execution flow

Adding the Jump Operation



Adding the Jump Operation



Mark active connections during execution flow

Instruction Times (Critical Paths)

- ❑ What is the clock cycle time assuming negligible delays for muxes, control unit, sign extend, PC access, shift left 2, wires, setup and hold times except:
 - ❑ Instruction and Data Memory (200 ps)
 - ❑ ALU and adders (200 ps)
 - ❑ Register File access (reads or writes) (100 ps)

Instr.	I Mem	Reg Rd	ALU Op	D Mem	Reg Wr	Total
R-type						
load						
store						
beq						
jump						

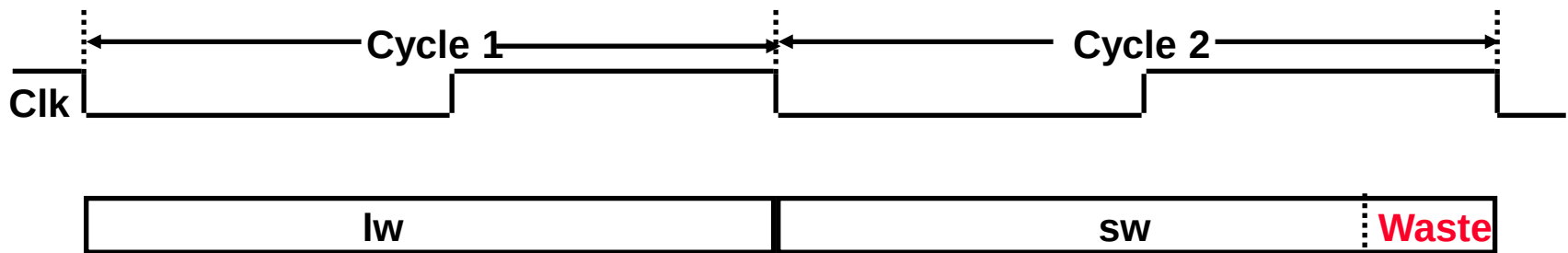
Instruction Critical Paths for Single cycle CPU

- ❑ What is the clock cycle time assuming negligible delays for muxes, control unit, sign extend, PC access, shift left 2, wires, setup and hold times except:
 - ❑ Instruction and Data Memory (200 ps)
 - ❑ ALU and adders (200 ps)
 - ❑ Register File access (reads or writes) (100 ps)

Instr.	I Mem	Reg Rd	ALU Op	D Mem	Reg Wr	Total
R-type	200	100	200		100	600
load	200	100	200	200	100	800
store	200	100	200	200		700
beq	200	100	200			500
jump	200					200

Single Cycle Disadvantages & Advantages

- ❑ Uses the clock cycle inefficiently – the clock cycle must be timed to accommodate the **slowest** instruction
 - ❑ especially problematic for more complex instructions like floating point multiply



- ❑ May be wasteful of area since some functional units (e.g., adders) must be duplicated since they can not be shared during a clock cycle

but

- ❑ Is simple and easy to understand

How Can We Make The Computer Faster?

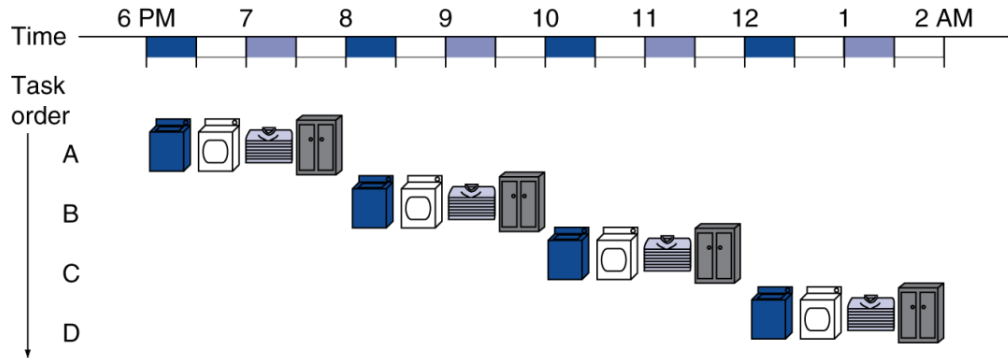
- ❑ Divide instruction cycles into smaller cycles
- ❑ Executing instructions in parallel
 - ❑ With only one CPU?
- ❑ Pipelining:
 - ❑ Start fetching and executing the next instruction before the current one has completed
 - ❑ Overlapping execution

Pipeline in real life



A more serious example: laundry work

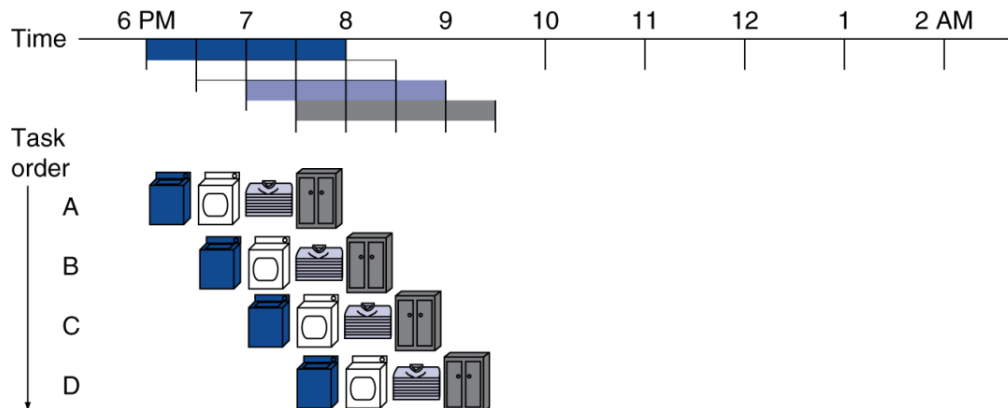
❑ Pipelined laundry boots performance up to 4 times



■ With 4 loads

$$T_{\text{normal}} = 4 \times 2 = 8 \text{ hours}$$

$$T_{\text{pipeline}} = 3.5 \text{ hours}$$



■ With n loads

$$T_{\text{normal}} = n \times 2 \text{ hours}$$

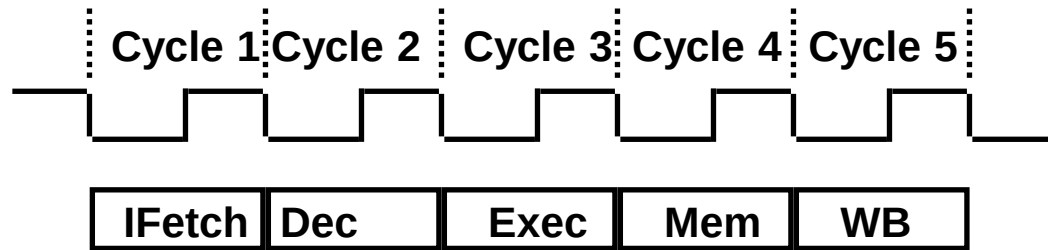
$$T_{\text{pipeline}} = (3+n)/2 \text{ hours}$$

4 stages: washing, drying, ironing, folding

When $n \rightarrow \infty$: $T_{\text{normal}} \rightarrow 4 \times T_{\text{pipeline}}$

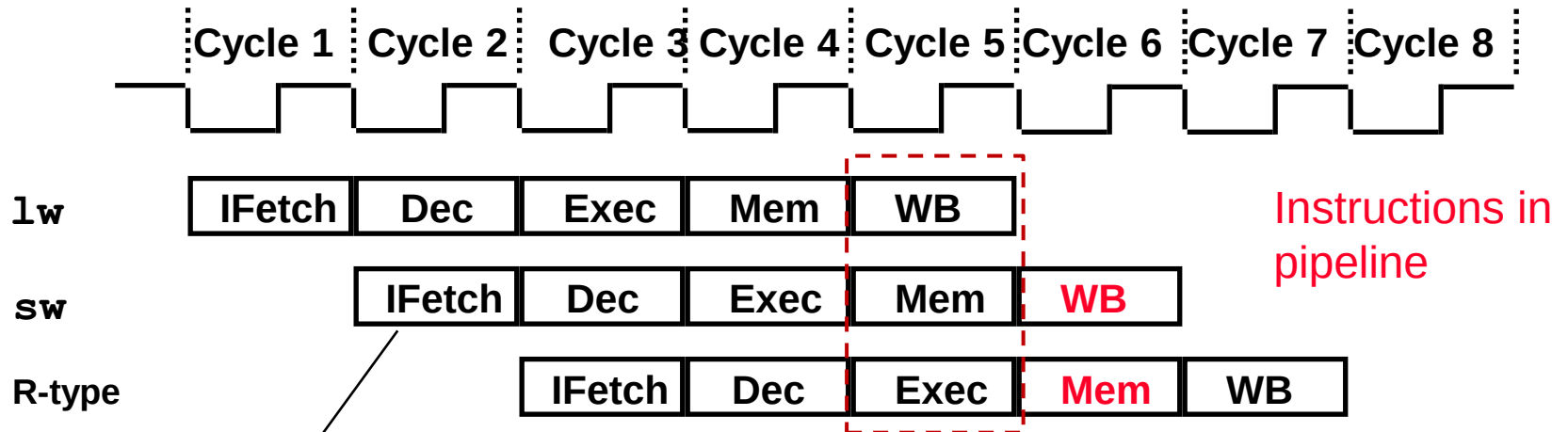
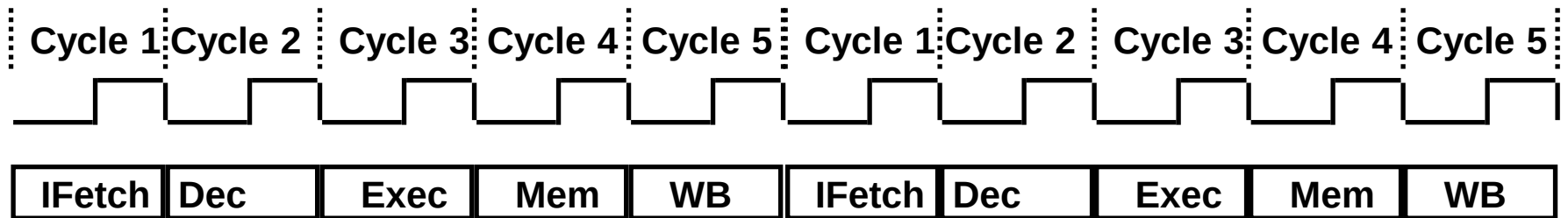
MIPS Pipeline

- ❑ Five stages, one step per stage
 - ❑ IFetch: Instruction Fetch and Update PC
 - ❑ Dec: Registers Fetch and Instruction Decode
 - ❑ Exec: Execute R-type; calculate memory address
 - ❑ Mem: Read/write the data from/to the Data Memory
 - ❑ WB: Write the result data into the register file



Execution time for a single instruction is always 5 cycles, regardless of instruction operation

Instruction pipeline

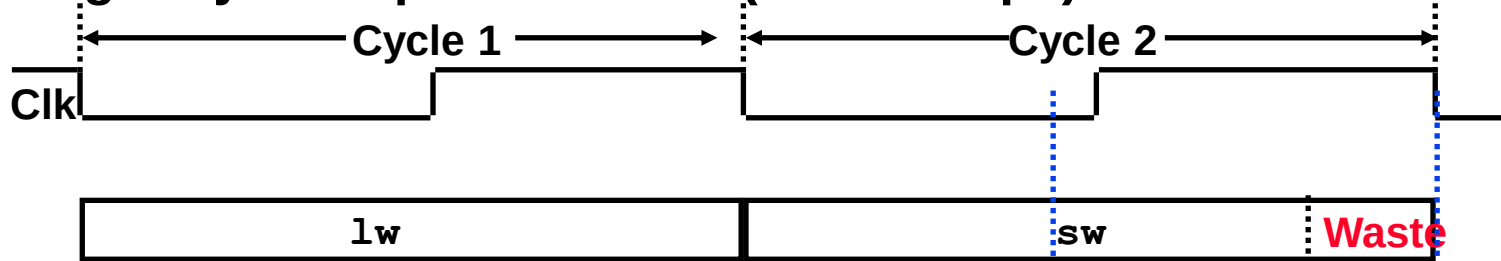


Start fetching and executing the next instruction before the current one has completed

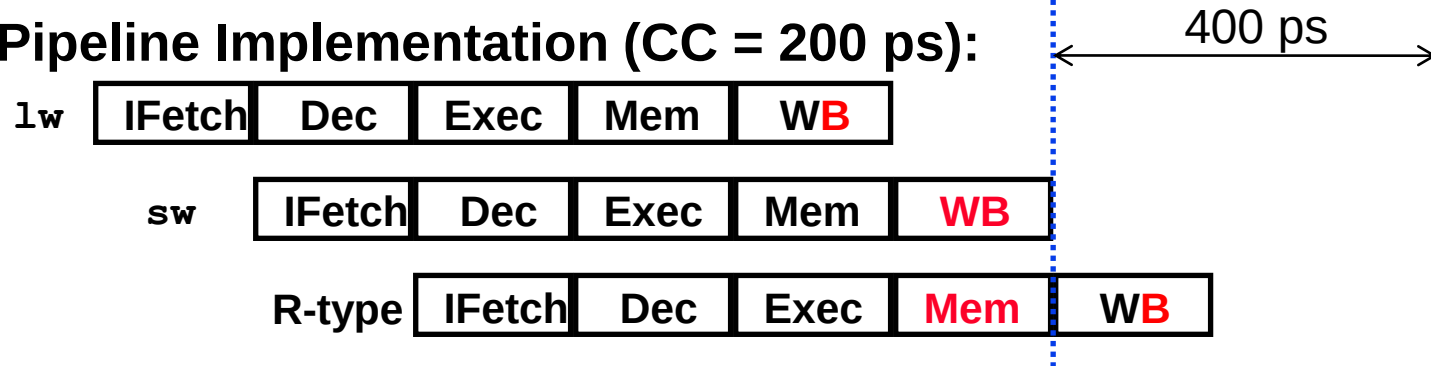
More than one instruction are executed at a time

Single Cycle versus Pipeline

Single Cycle Implementation (CC = 800 ps):

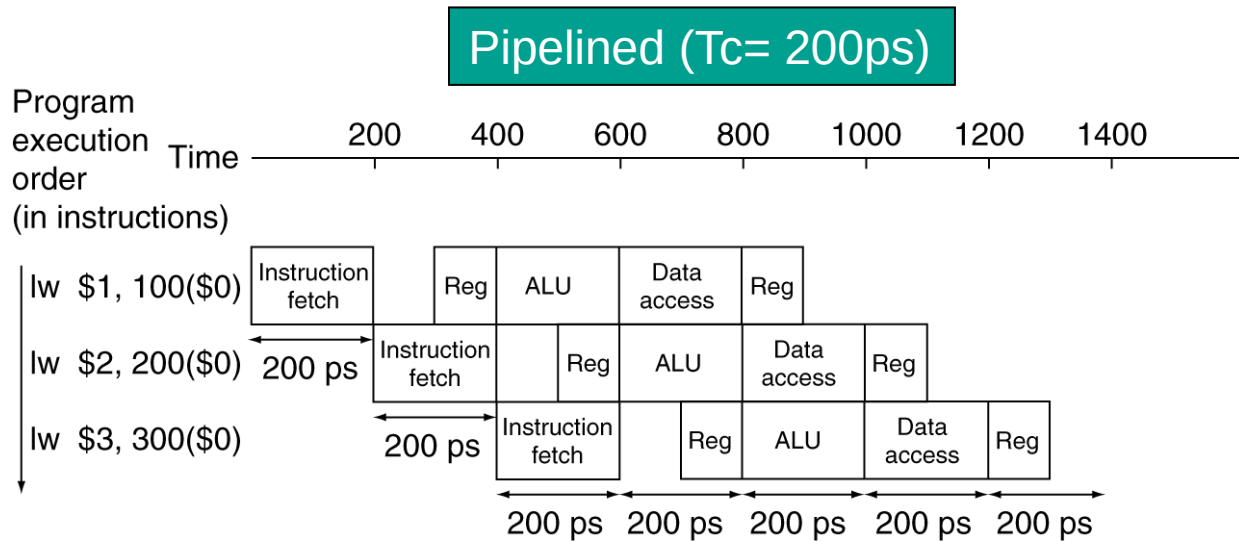
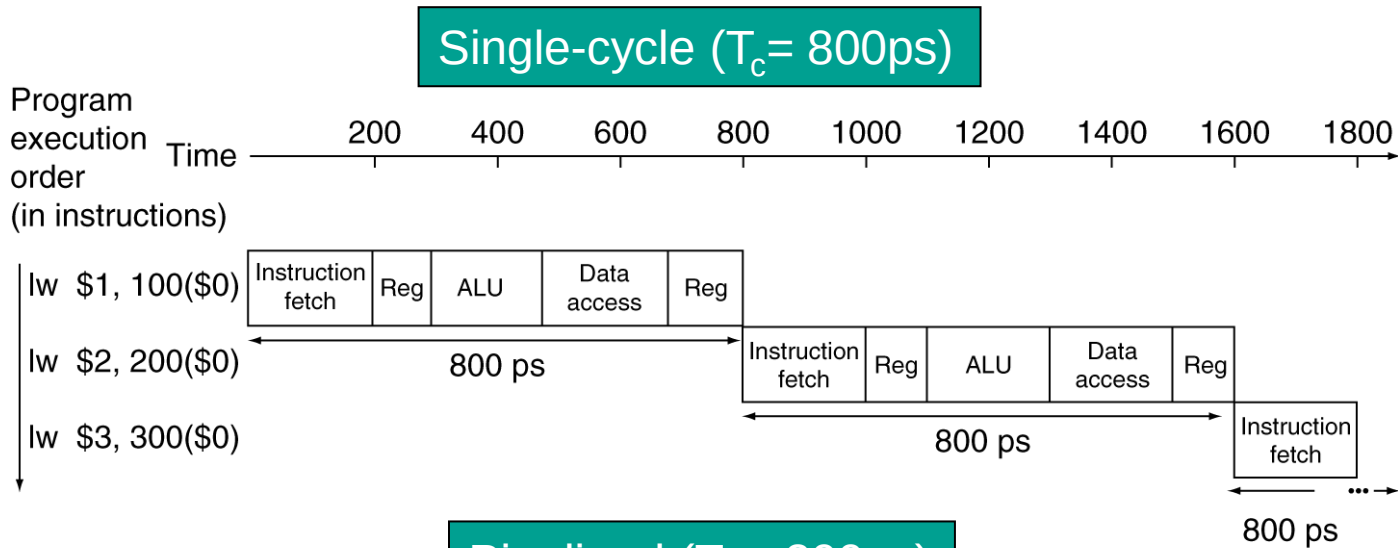


Pipeline Implementation (CC = 200 ps):



- ❑ To complete an entire instruction in the pipelined case takes 1000 ps (as compared to 800 ps for the single cycle case). Why ?
- ❑ How long does each take to complete 1,000,000 adds ?

Example with lw instructions



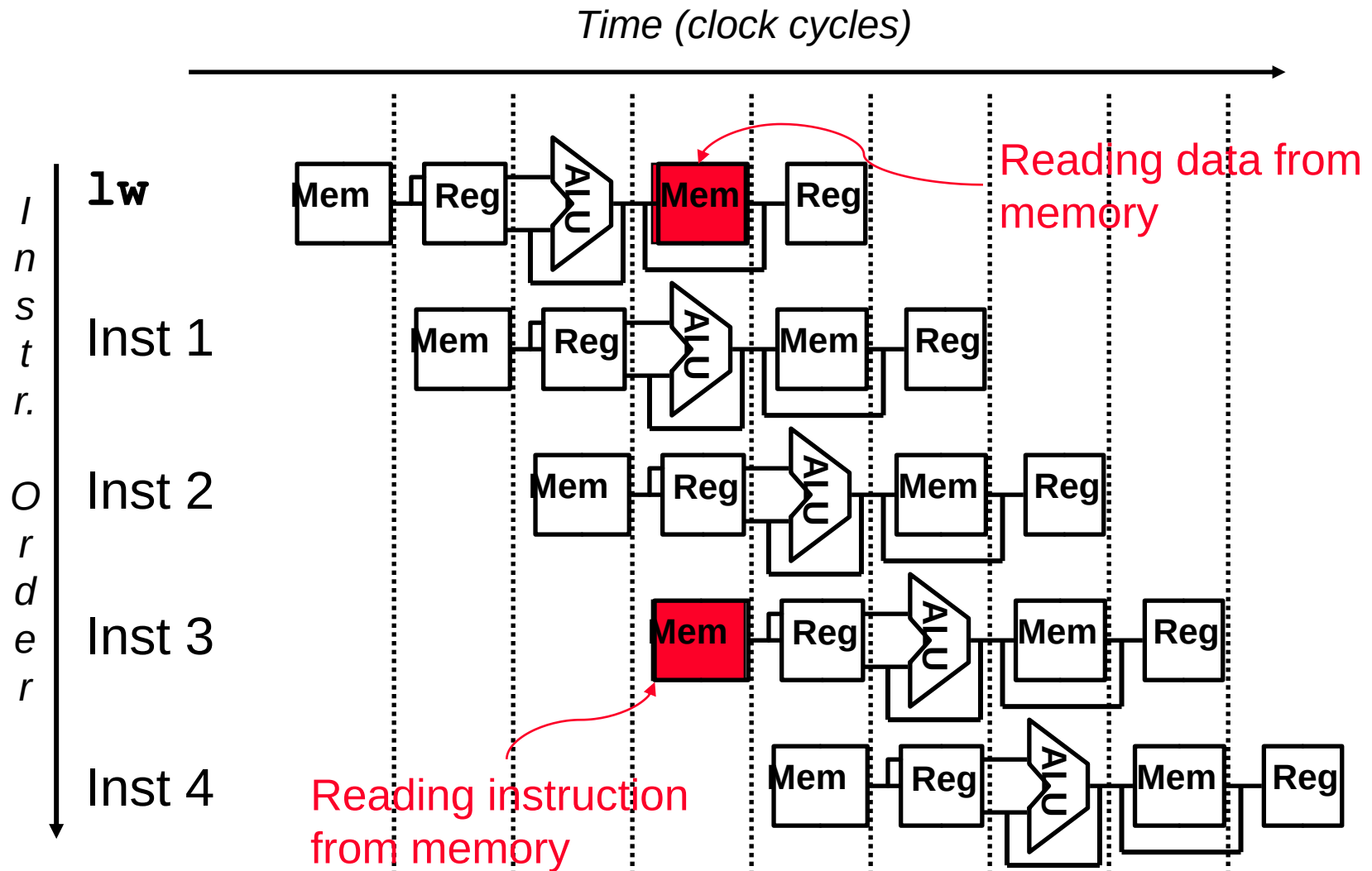
Pipeline hazards

- ❑ Pipeline can lead us into troubles!!!
- ❑ Hazards: situations that prevent starting the next instruction in the next cycle
 - ❑ **structural hazards**: attempt to use the same resource by two different instructions at the same time
 - ❑ **data hazards**: attempt to use data before it is ready
 - An instruction's source operand(s) are produced by a prior instruction still in the pipeline
 - ❑ **control hazards**: attempt to make a decision about program control flow before the condition has been evaluated and the new PC target address calculated
 - branch and jump instructions, exceptions
- ❑ In most cases, hazard can be solved simply by waiting
 - ❑ but we need better solutions to take advantages of pipeline

Structural hazard

- ❑ Conflict for use of a resource
- ❑ In MIPS pipeline with a single memory
 - ❑ Load/store requires data access
 - ❑ Instruction fetch would have to *stall* for that cycle
 - Would cause a pipeline “bubble”
- ❑ Hence, pipelined datapath require separate instruction/data memories
 - ❑ Or separate instruction/data caches

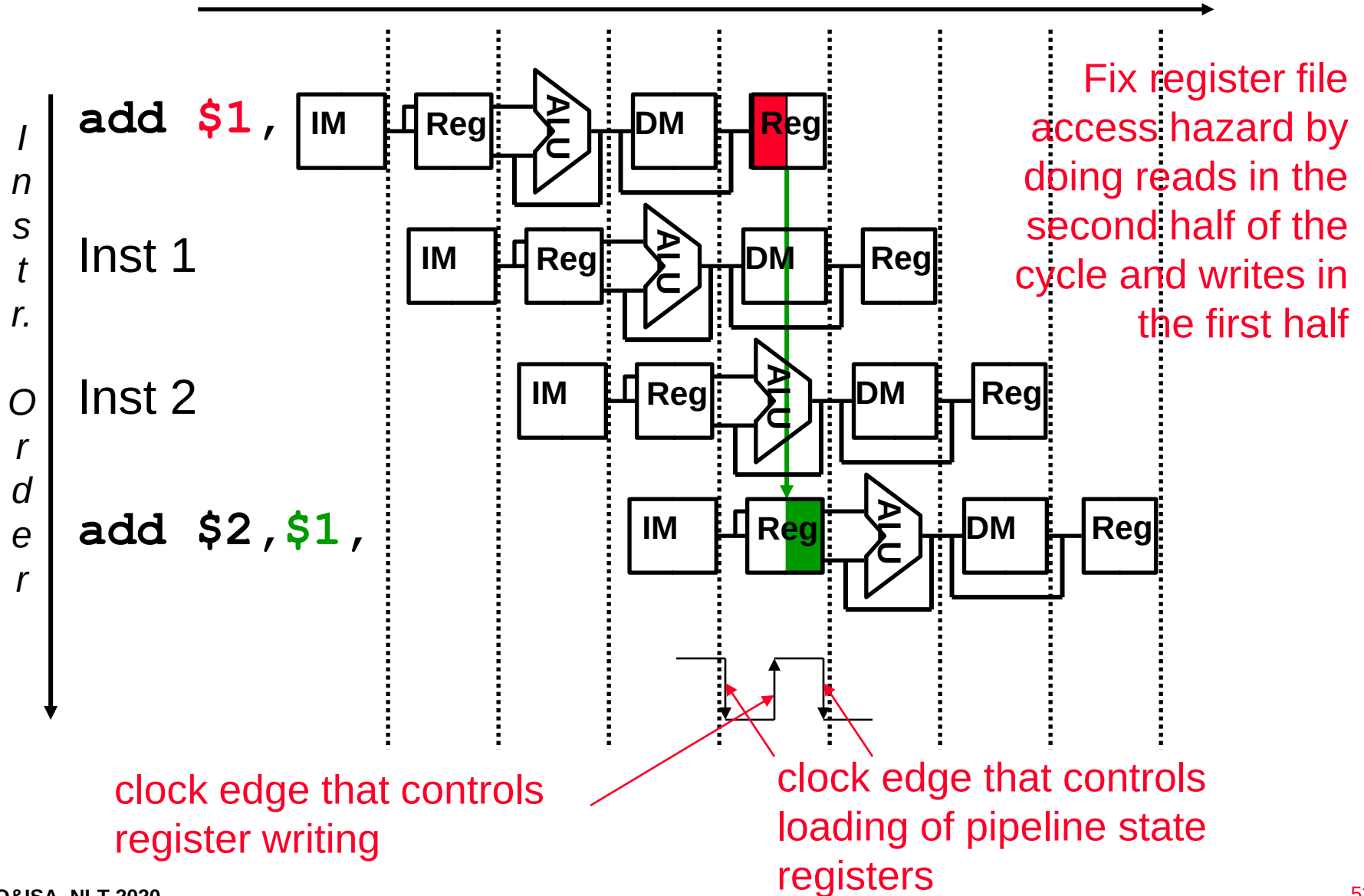
A Single Memory Would Be a Structural Hazard



❑ Fix with separate instr and data memories (I\$ and D\$)

How About Register File Access?

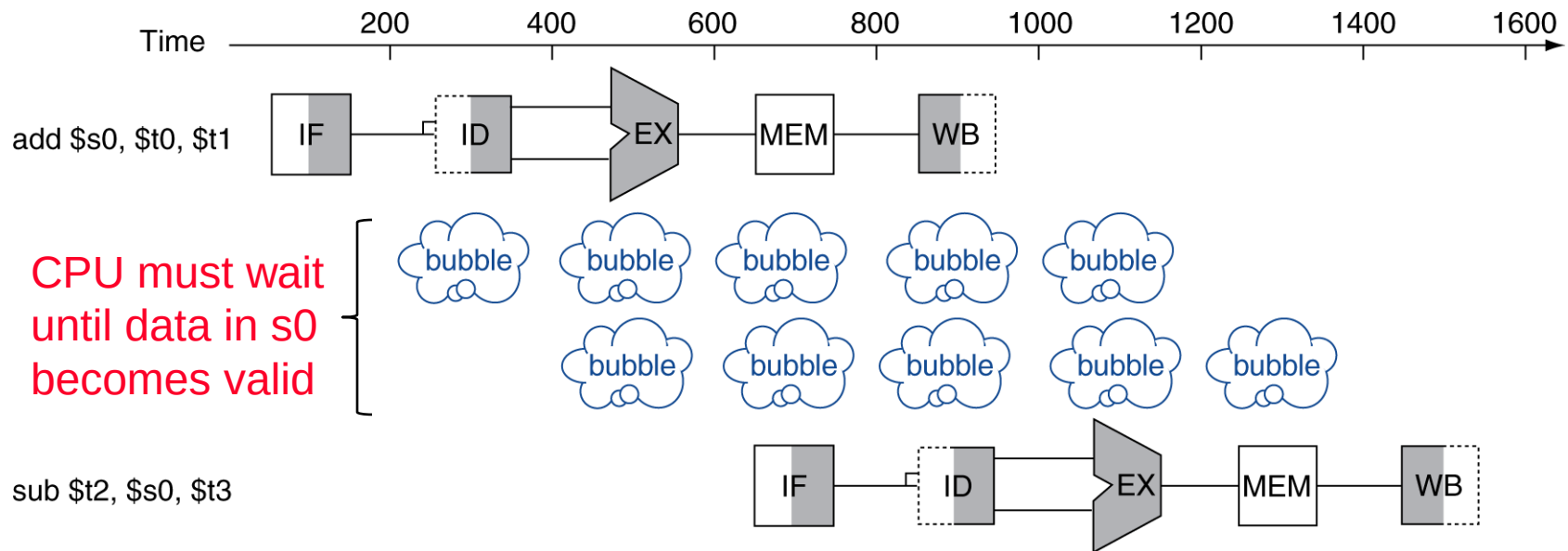
Time (clock cycles)



Data hazard

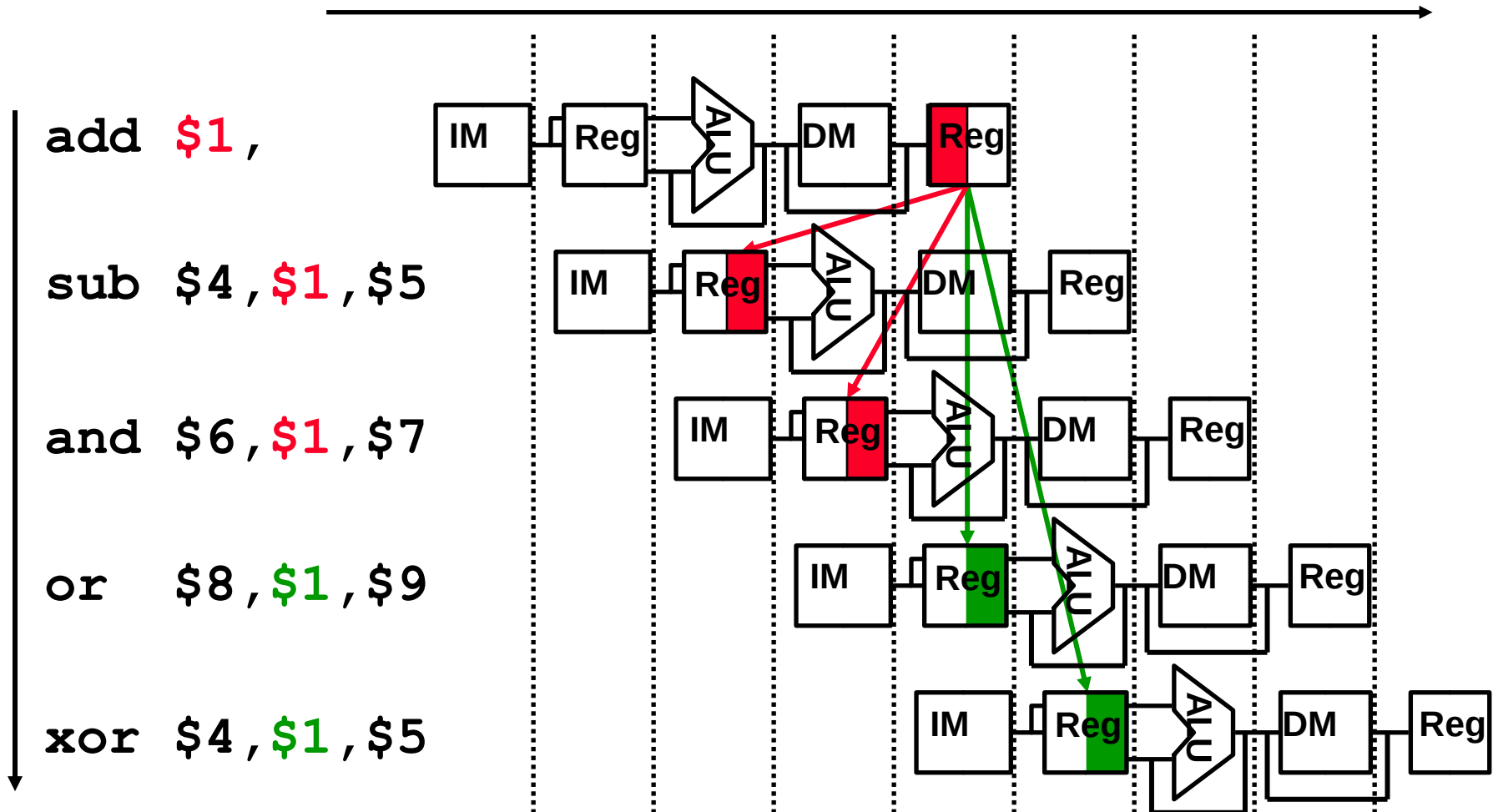
❑ An instruction depends on completion of data access by a previous instruction

❑ add \$s0, \$t0, \$t1
 sub \$t2, \$s0, \$t3



Example

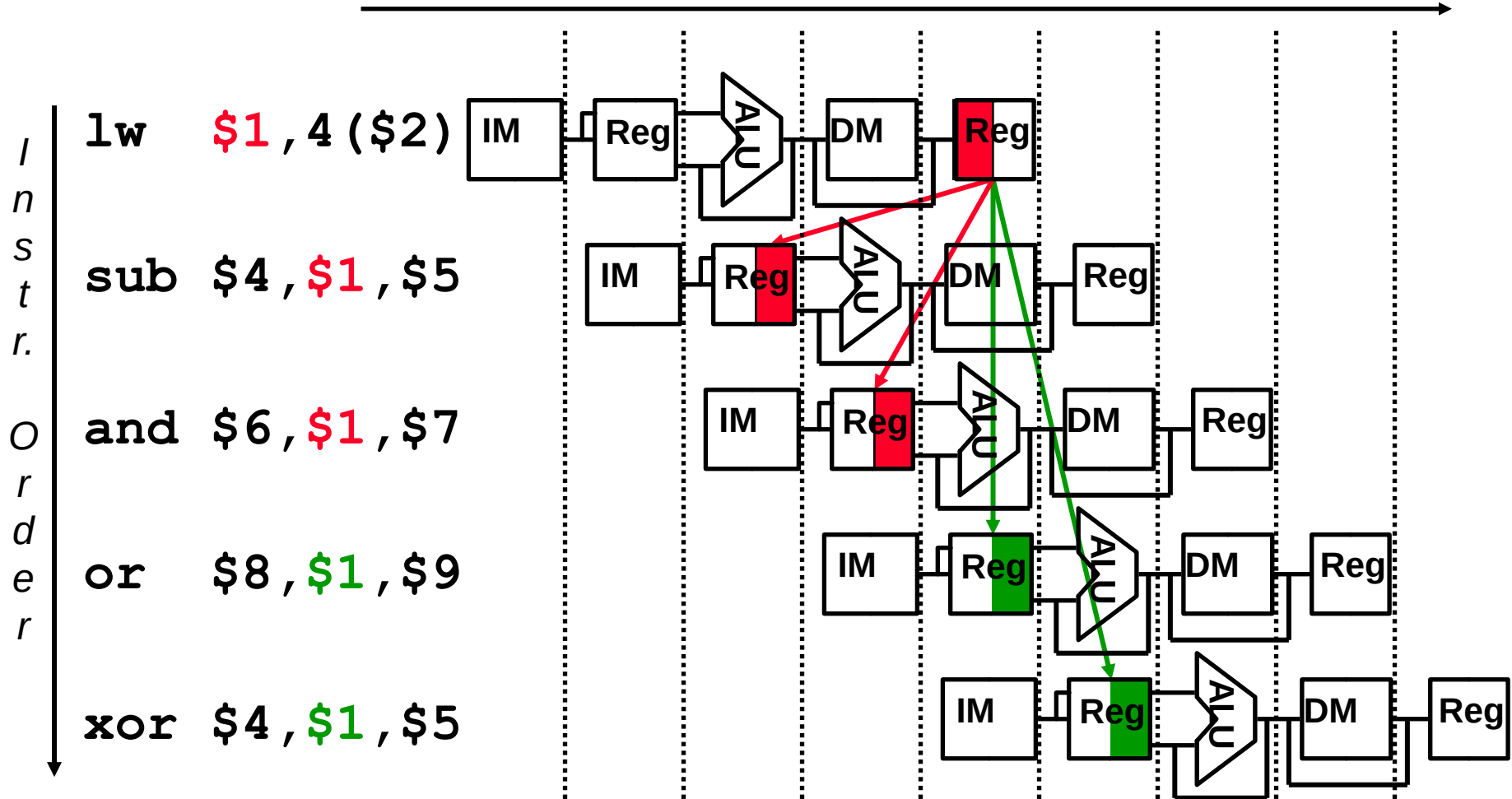
- Dependencies backward in time cause **hazards**



- Read before write data hazard**

Example

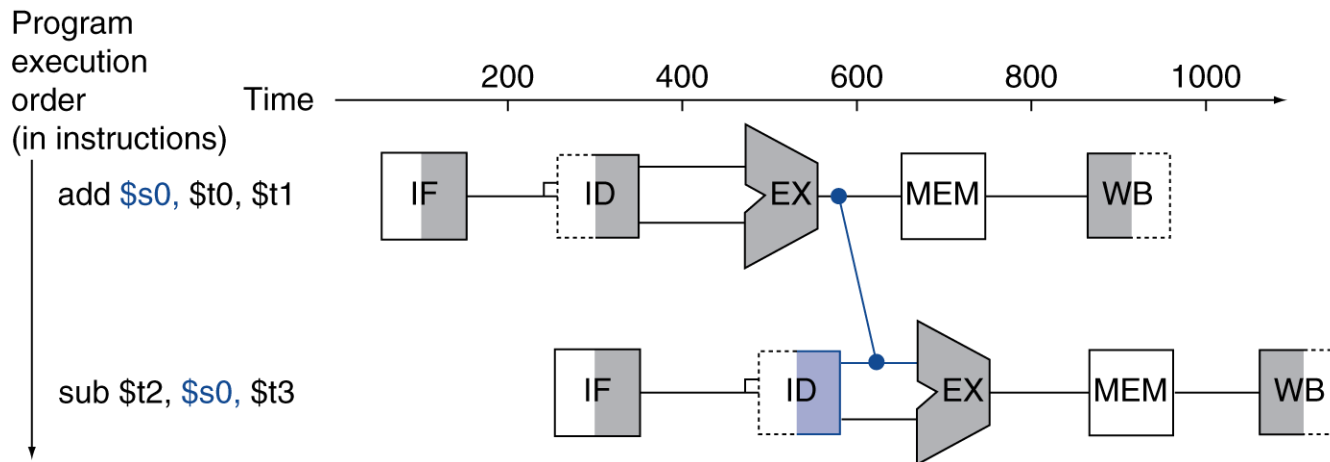
❑ Dependencies backward in time cause **hazards**



❑ Load-use data hazard

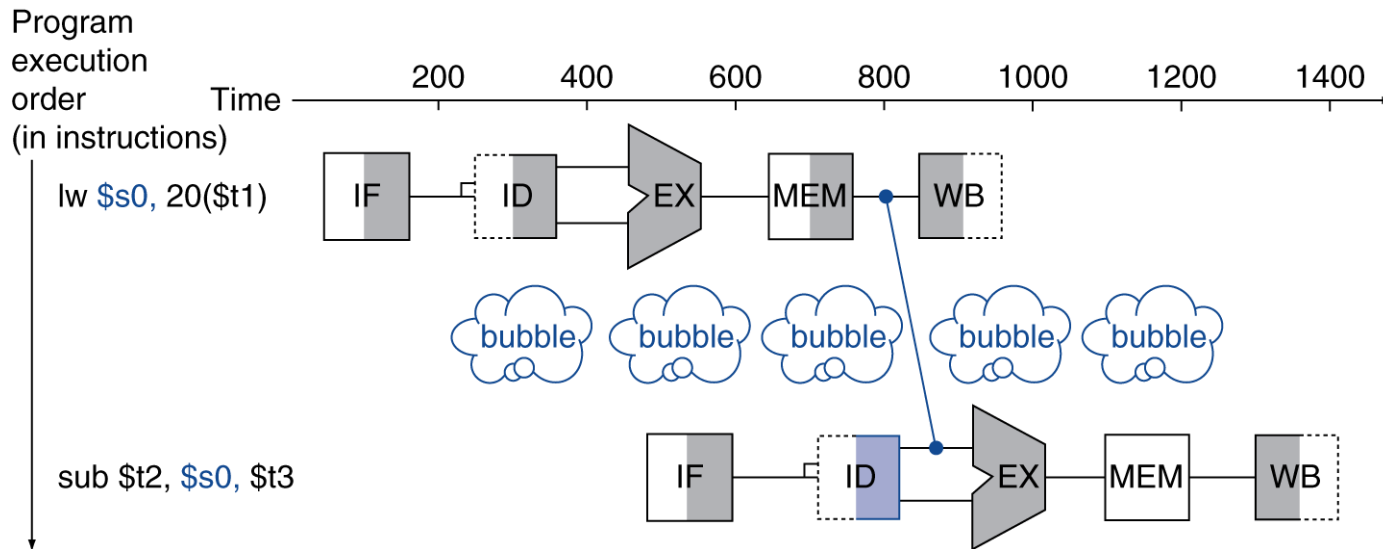
Solving hazard with forwarding

- ❑ Use result when it is computed
 - ❑ Don't wait for it to be stored in a register
 - ❑ Requires extra connections in the datapath
- ❑ Forward from EX to EX (output to input)



Load-Use Data Hazard

- ❑ One cycle stall is necessary
- ❑ Forward from MEM (output) to EX (input)

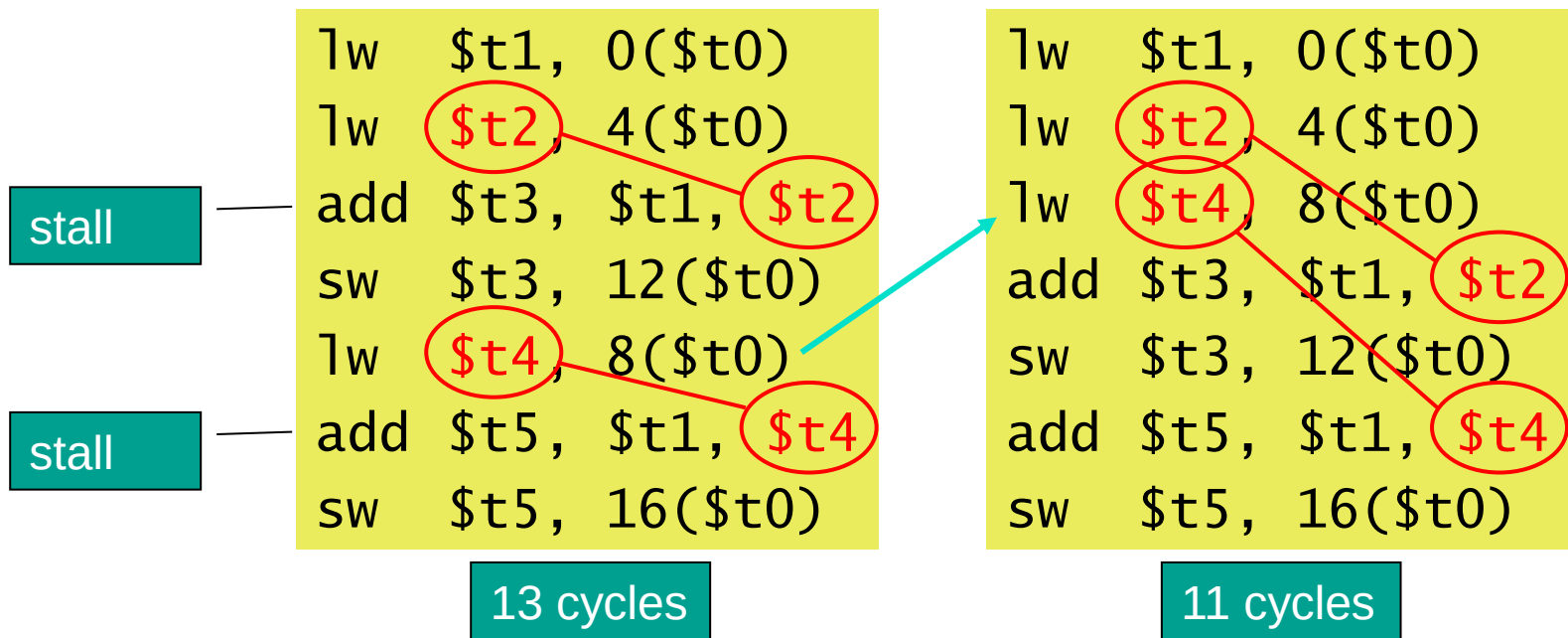


Code Scheduling to Avoid Stalls

- Reorder code to avoid use of load result in the next instruction

C code: $A = B + E;$

$C = B + F;$

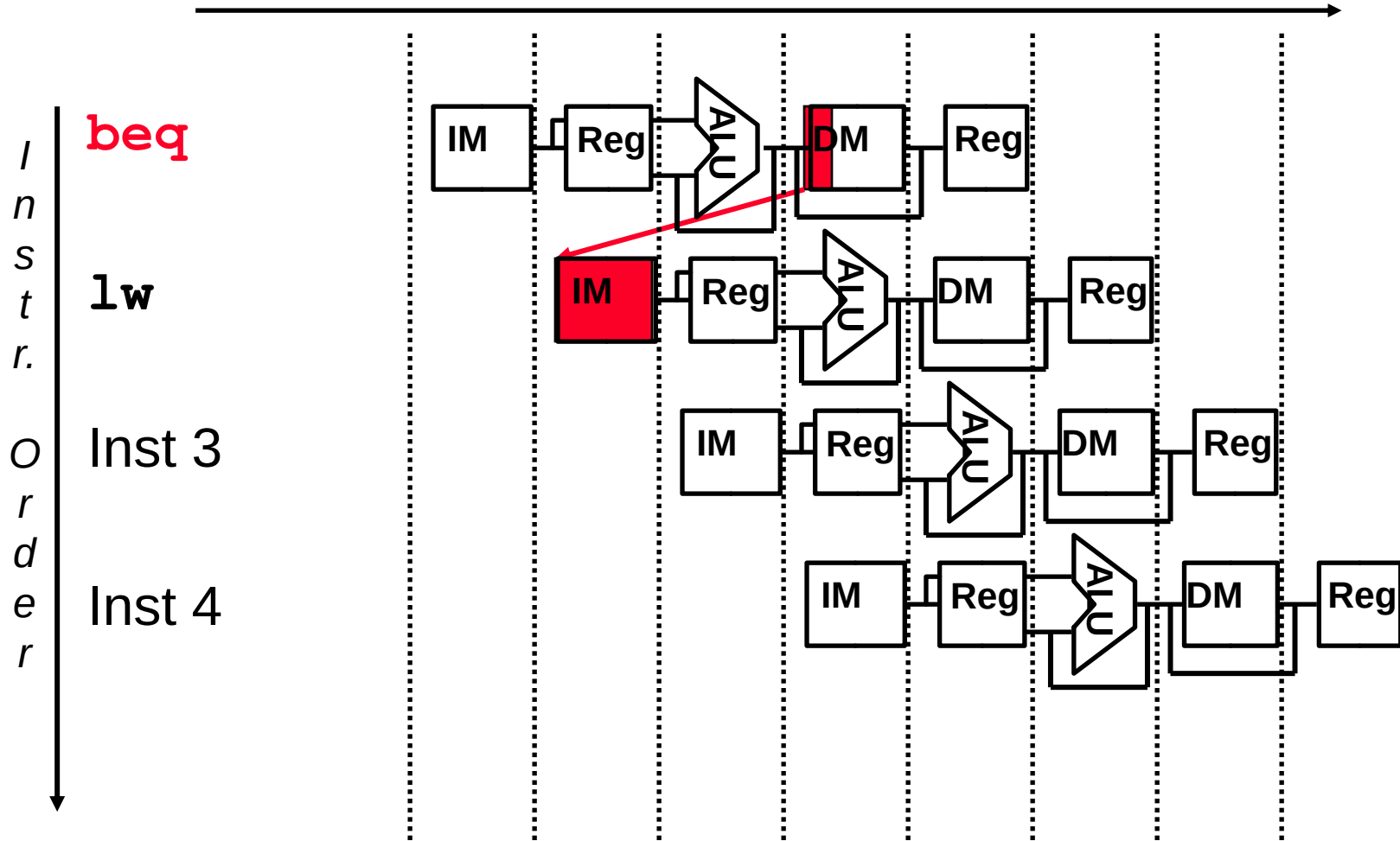


Control Hazards

- ❑ Branch determines flow of control
 - ❑ Fetching next instruction depends on branch outcome
 - ❑ Pipeline can't always fetch correct instruction
 - Still working on ID stage of branch
- ❑ In MIPS pipeline
 - ❑ Need to compare registers and compute target early in the pipeline
 - ❑ Add hardware to do it in ID stage

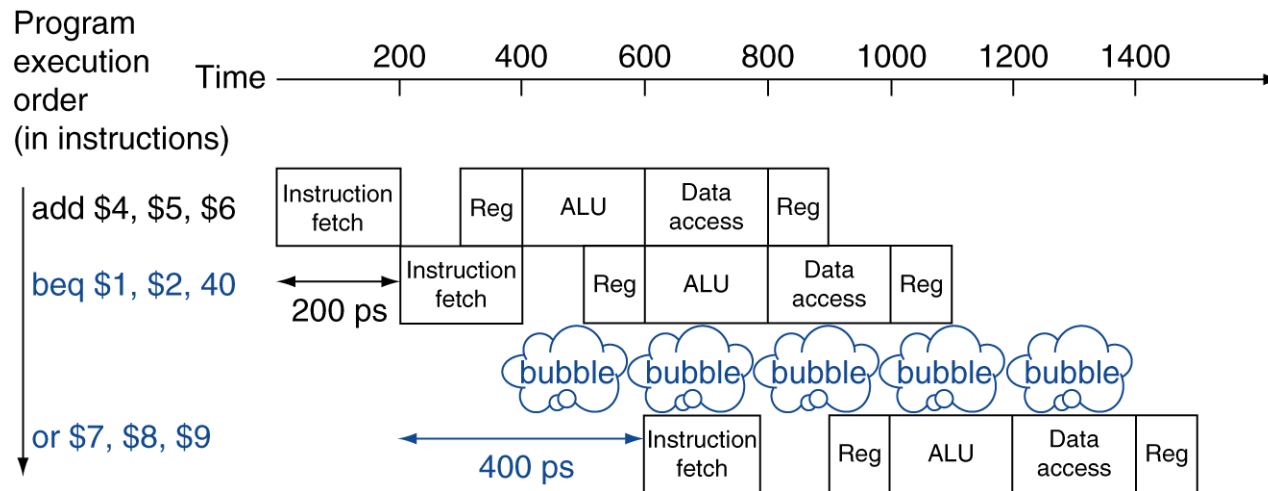
Branch Instructions Cause Control Hazards

- ❑ Dependencies backward in time cause **hazards**



Stall on Branch

- ❑ Naïve approach: Wait until branch outcome determined before fetching next instruction



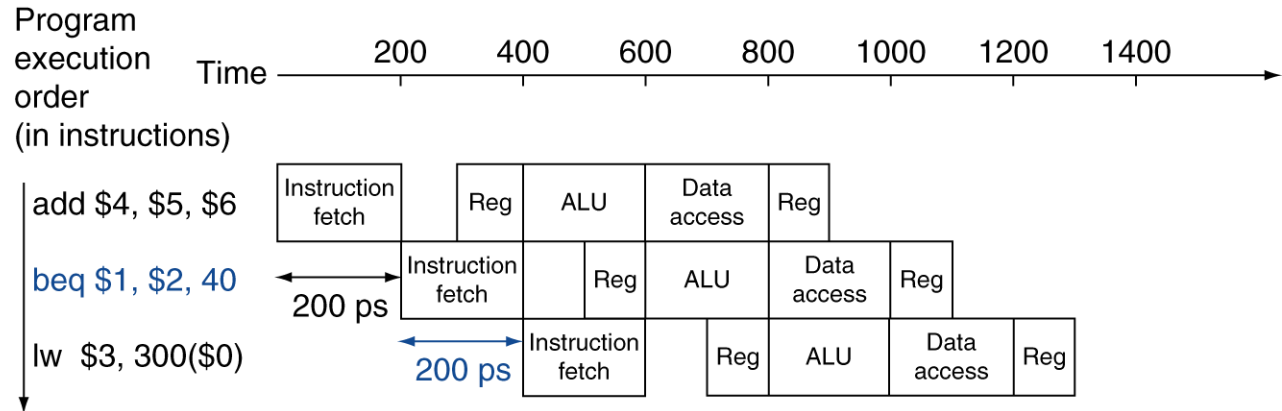
Performance affect: assume that 17% of instructions in program are branches, if each branch take one cycle for the stall, then performance will be 17% slower. (CPI = 1.17)

Branch Prediction

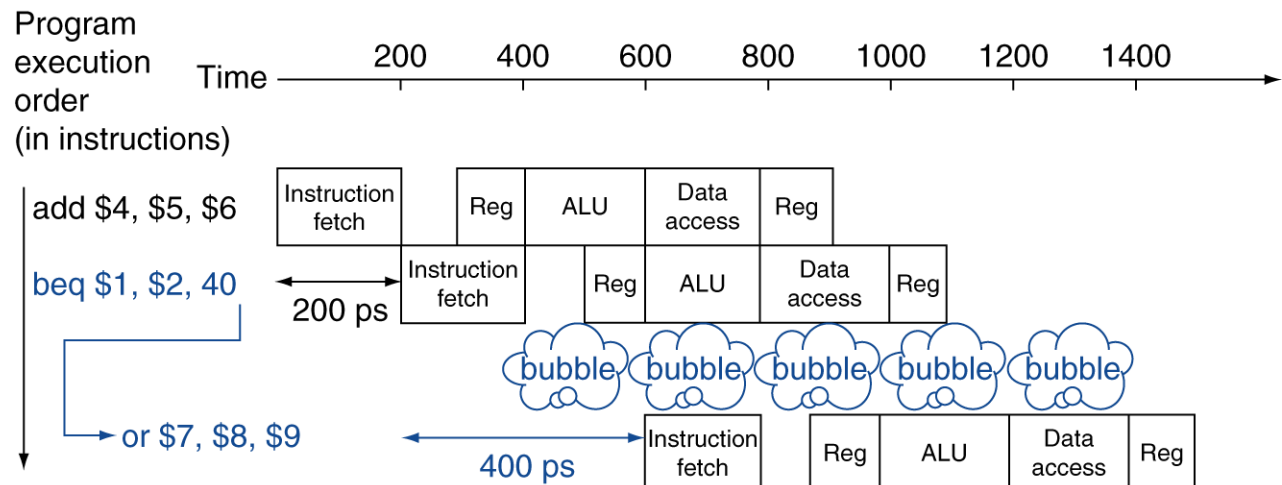
- ❑ Predict outcome of branch
- ❑ Only stall if prediction is wrong
- ❑ In MIPS pipeline
 - ❑ Can predict branches not taken
 - ❑ Fetch instruction after branch, with no delay

MIPS with Predict Not Taken

Prediction correct



Prediction incorrect



More-Realistic Branch Prediction

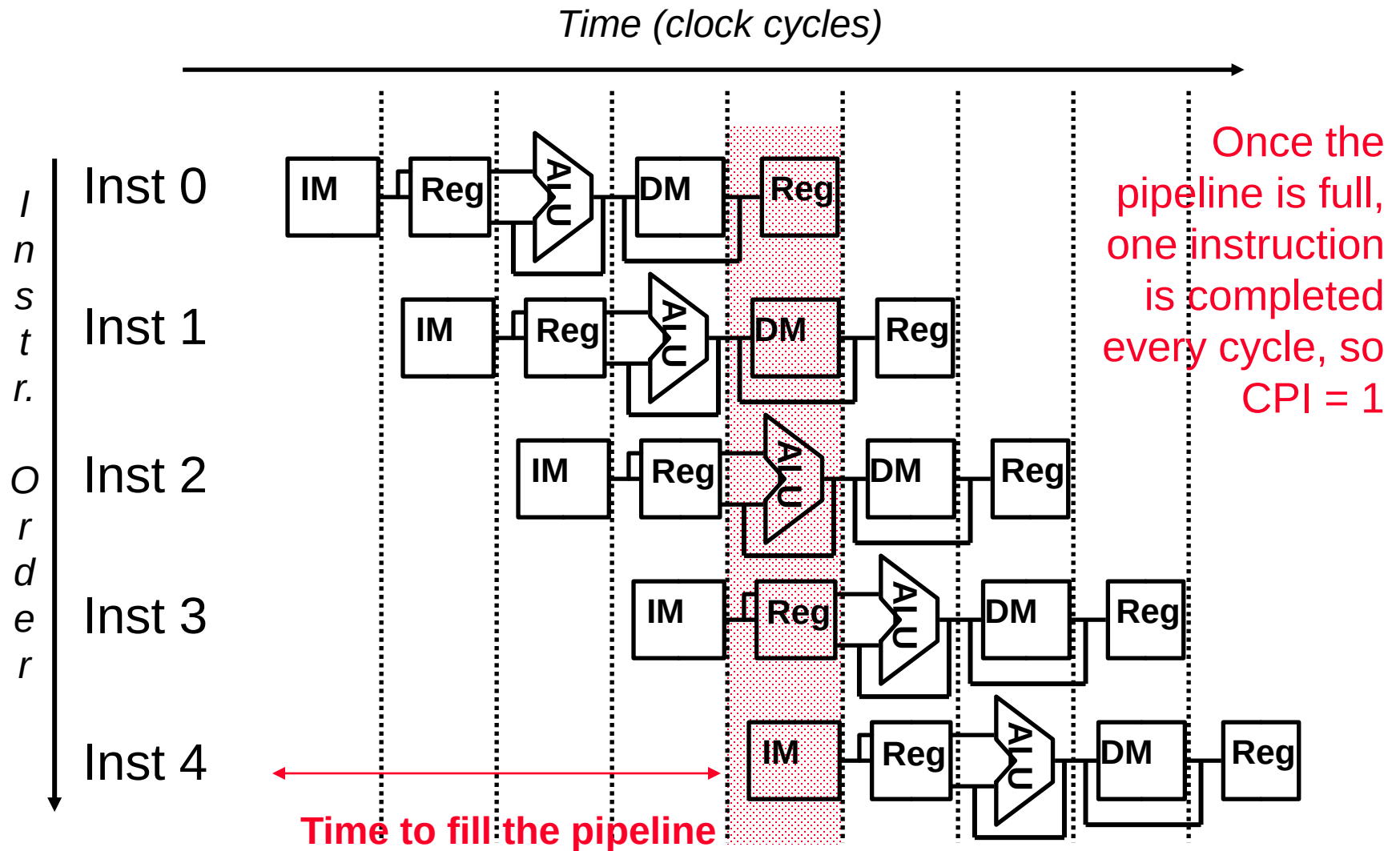
❑ Static branch prediction

- ❑ Based on typical branch behavior
- ❑ Example: loop and if-statement branches
 - Predict backward branches taken
 - Predict forward branches not taken

❑ Dynamic branch prediction

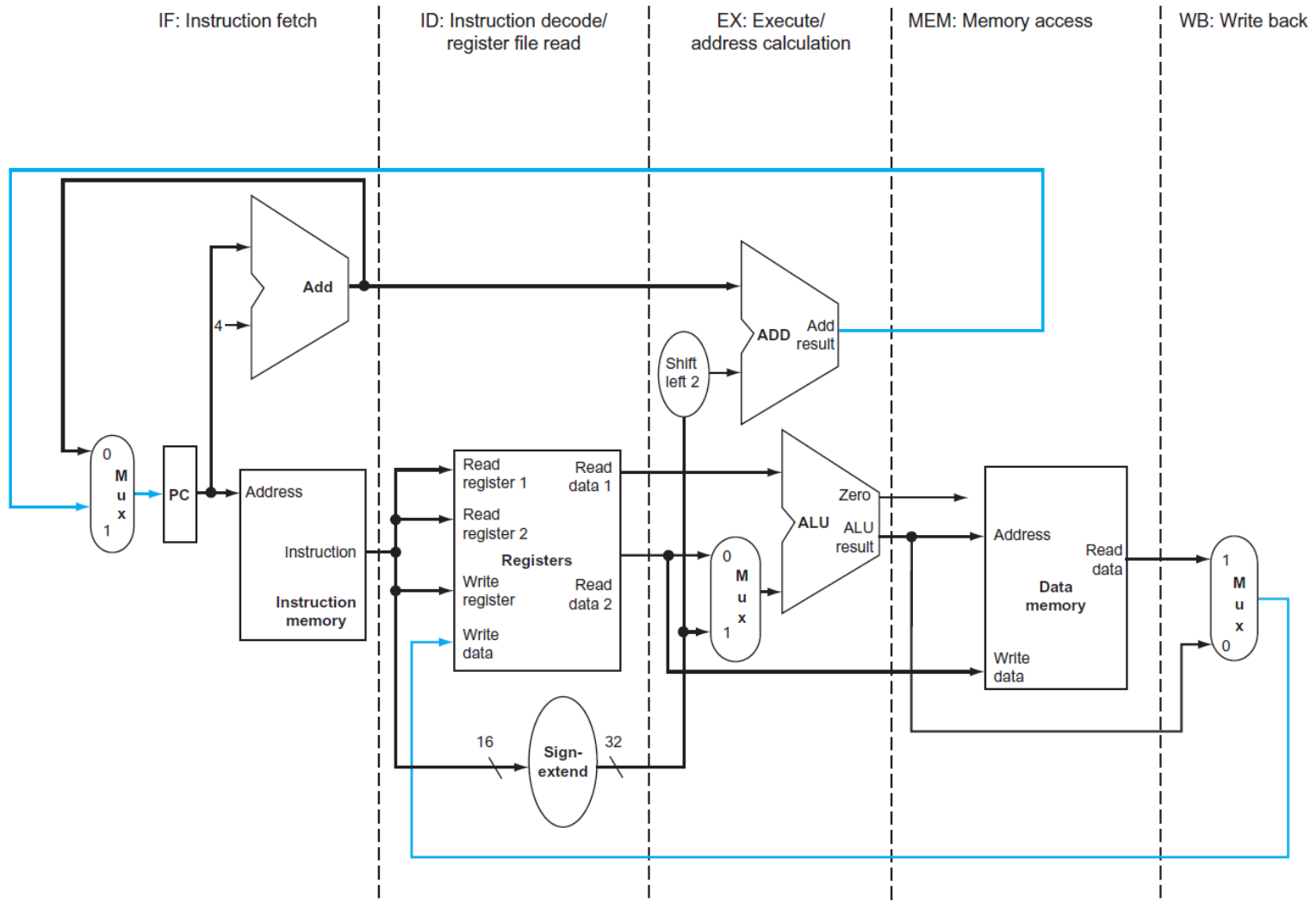
- ❑ Hardware measures actual branch behavior
 - e.g., record recent history of each branch
- ❑ Assume future behavior will continue the trend
 - When wrong, stall while re-fetching, and update history
- ❑ As good as $> 90\%$ accuracy

Summary: Pipeline Operation



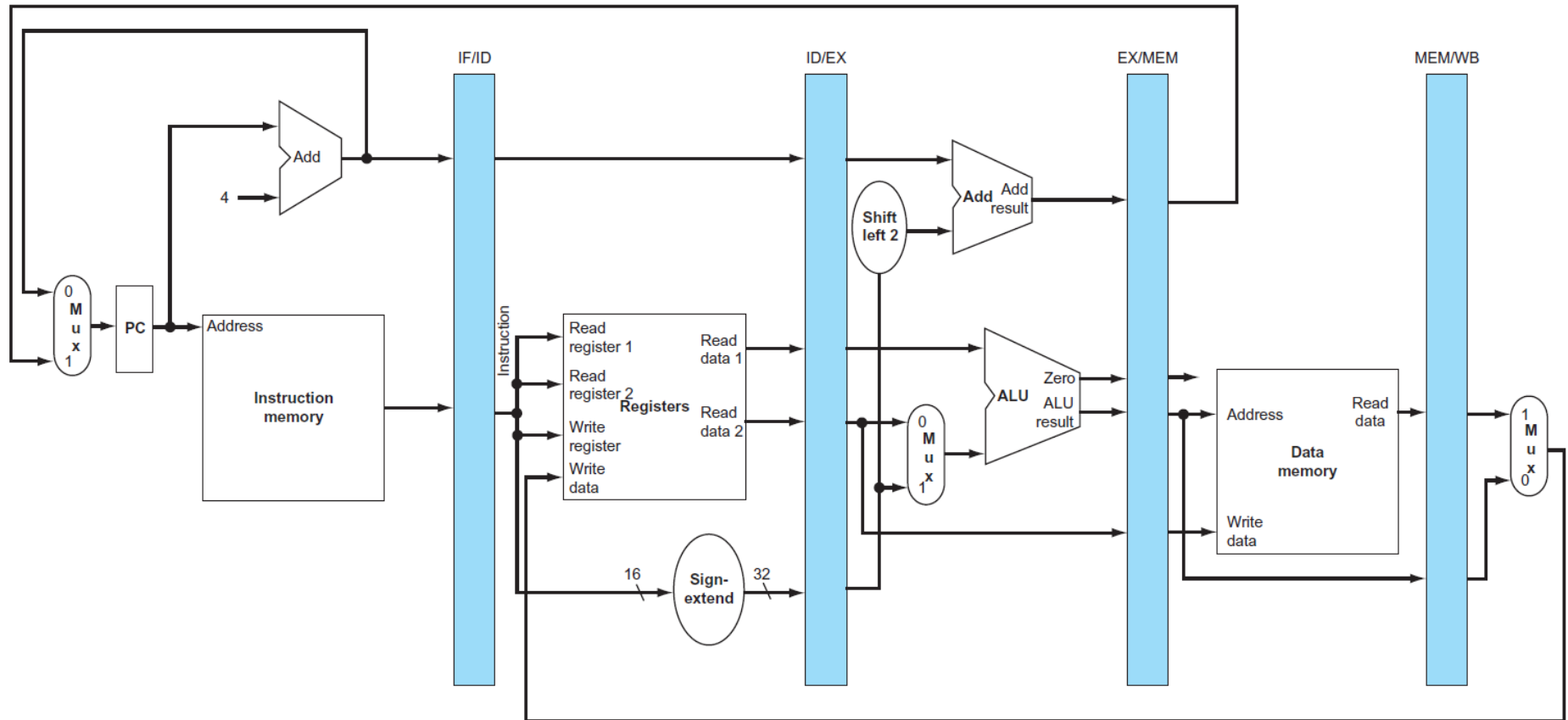
Pipelined datapath

❑ How to share/isolate data between different stages?



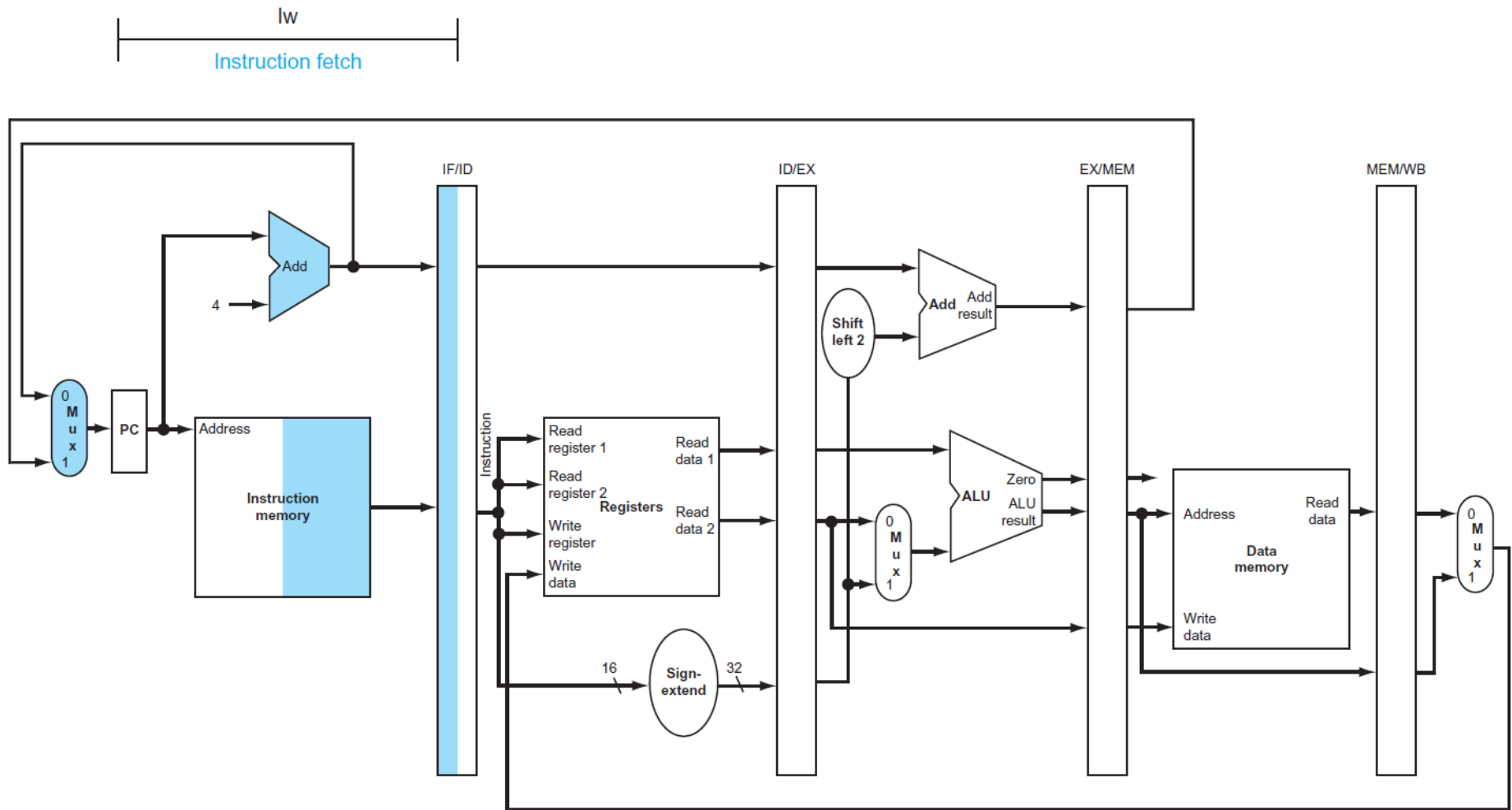
Pipelined datapath

❑ Adding pipeline registers

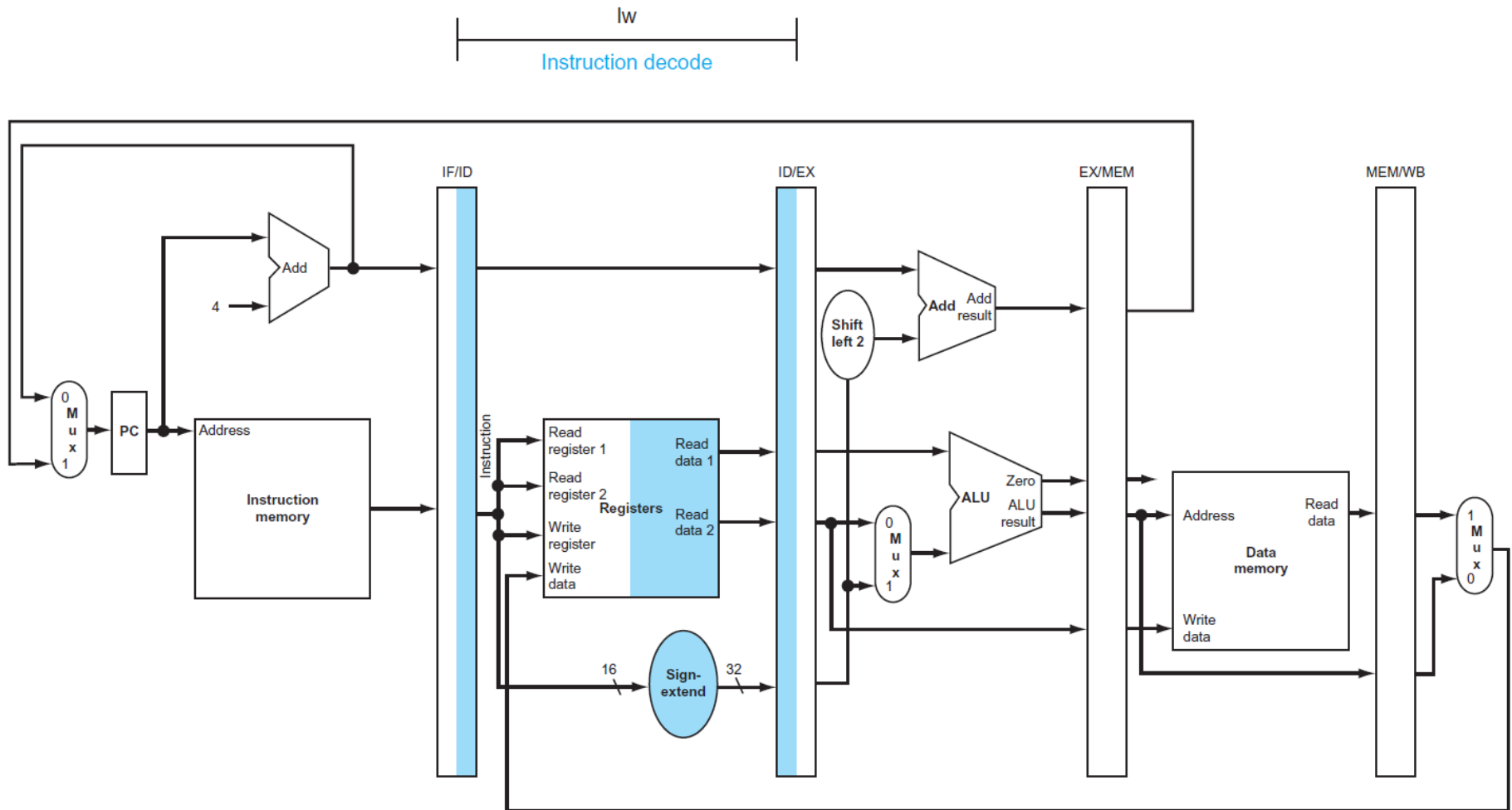


❑ 5 stages, why only 4 registers?

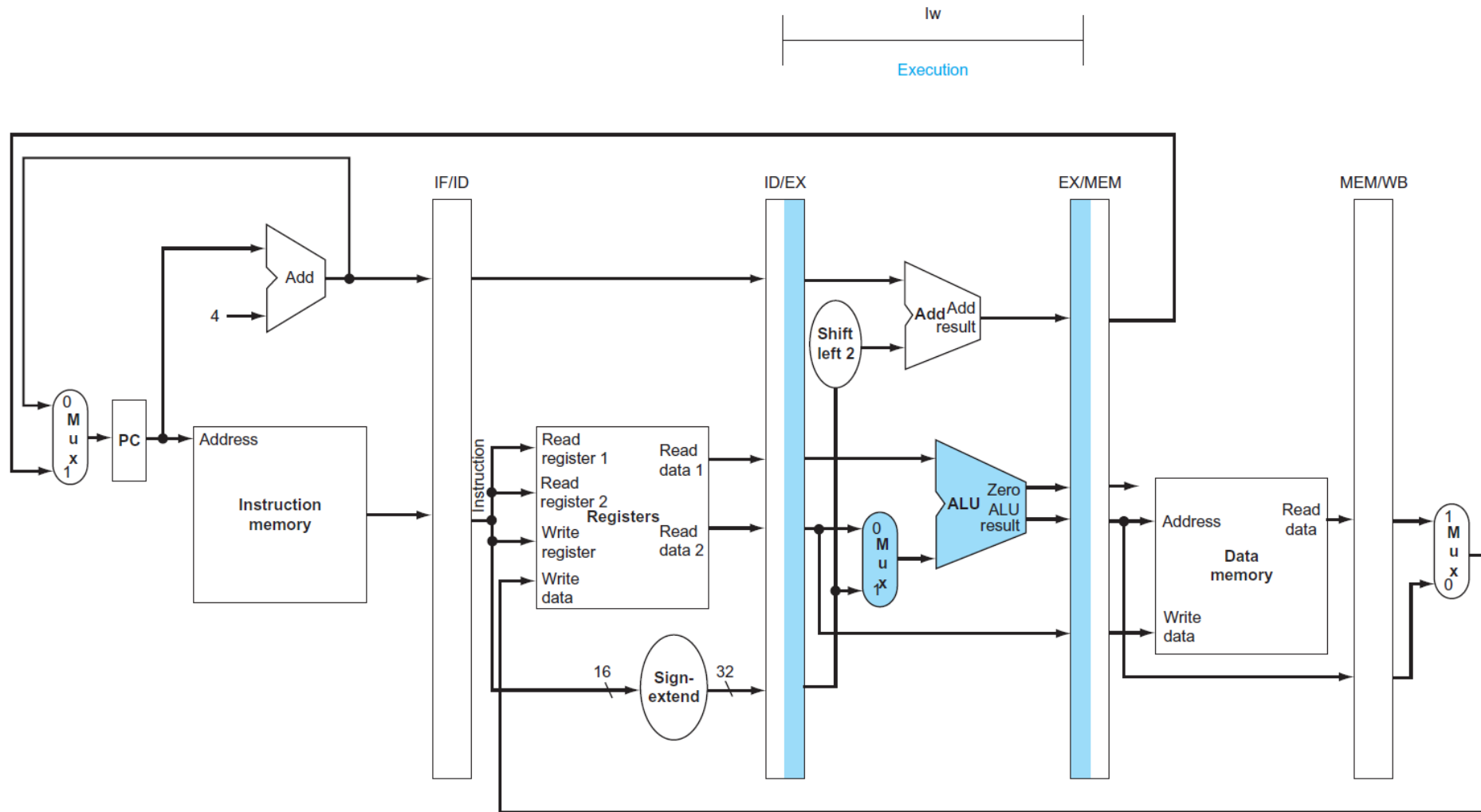
Instruction fetch in pipeline



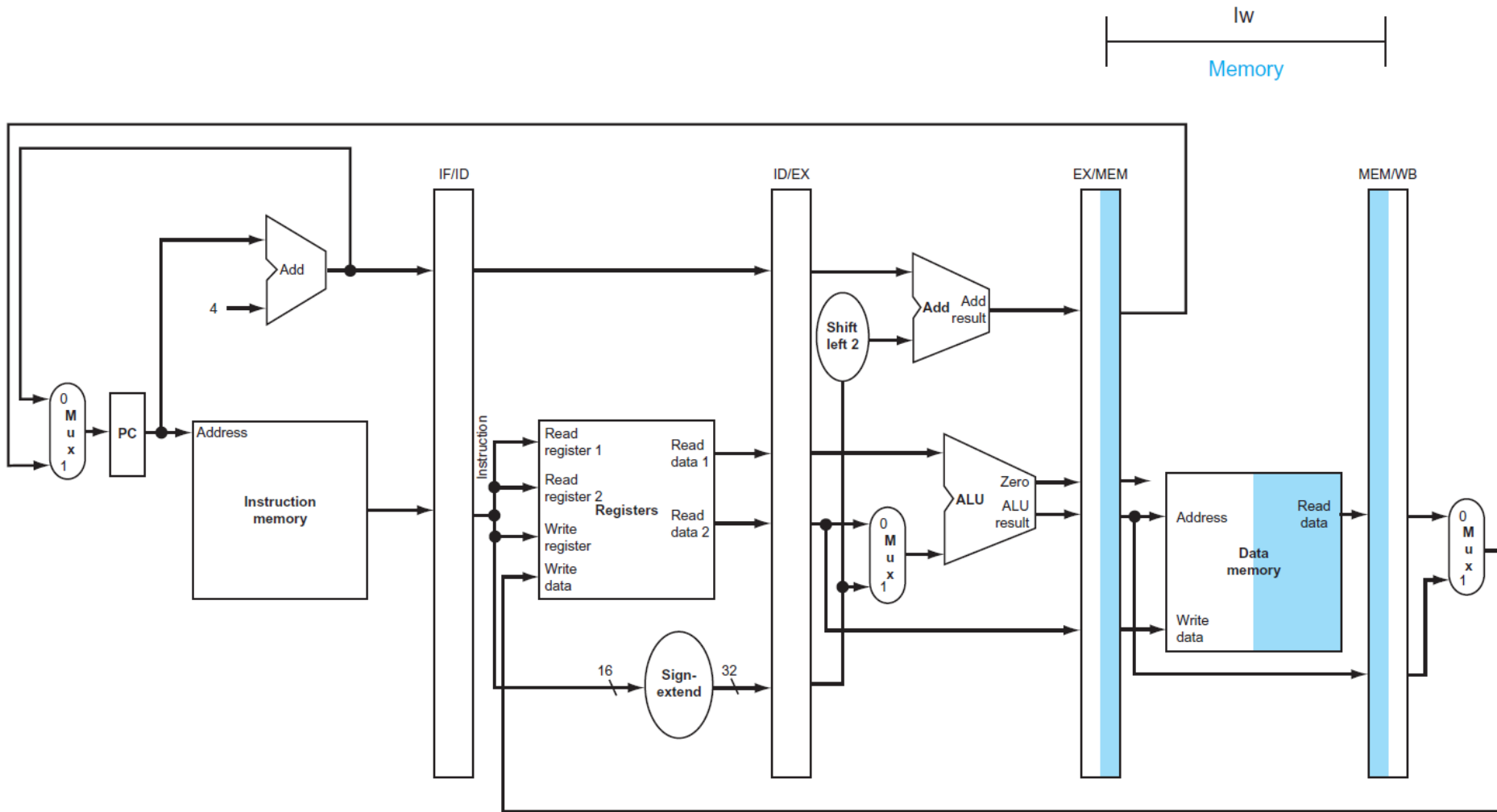
Instruction decode



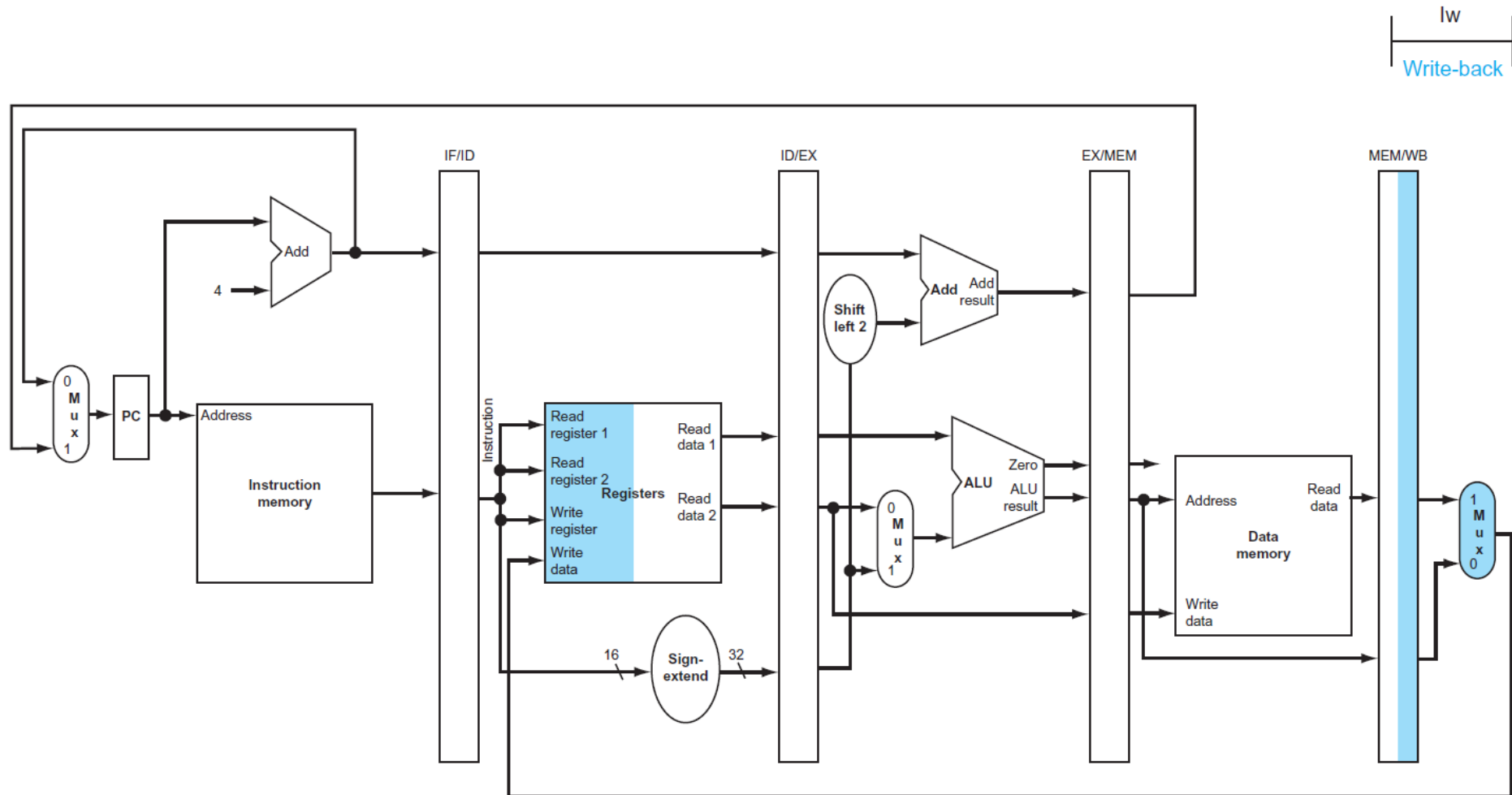
Execution



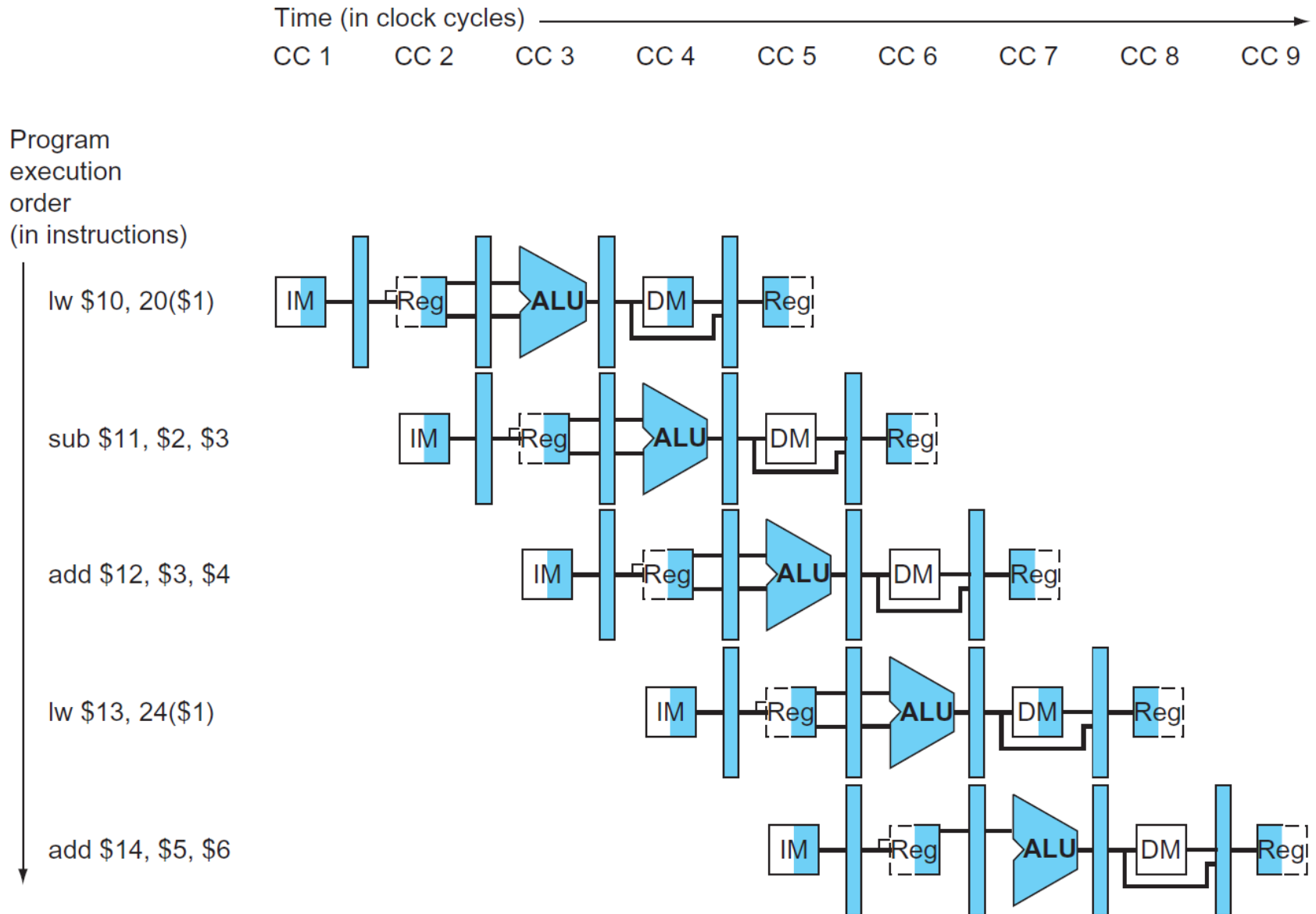
Memory access



Write-back

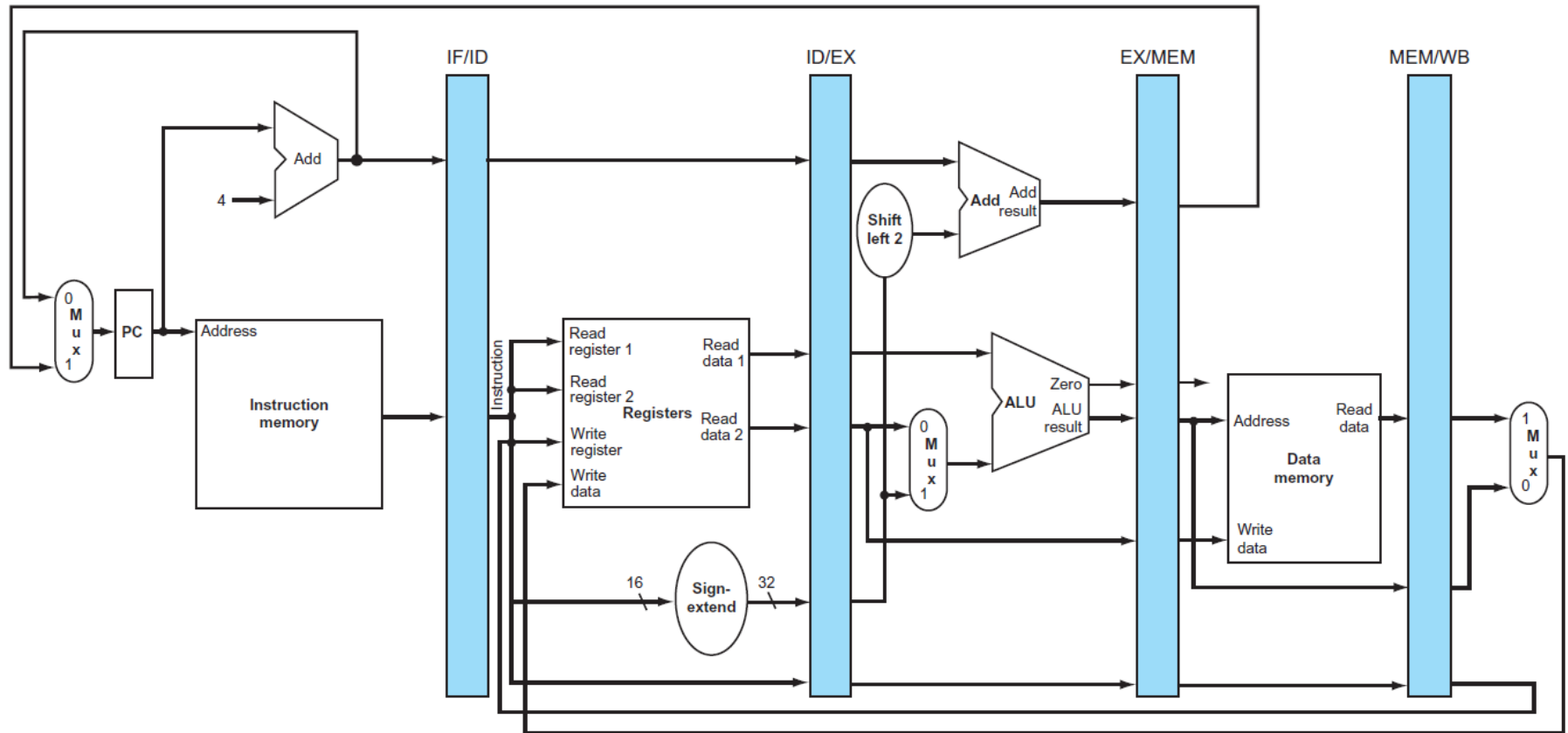


Pipeline diagram

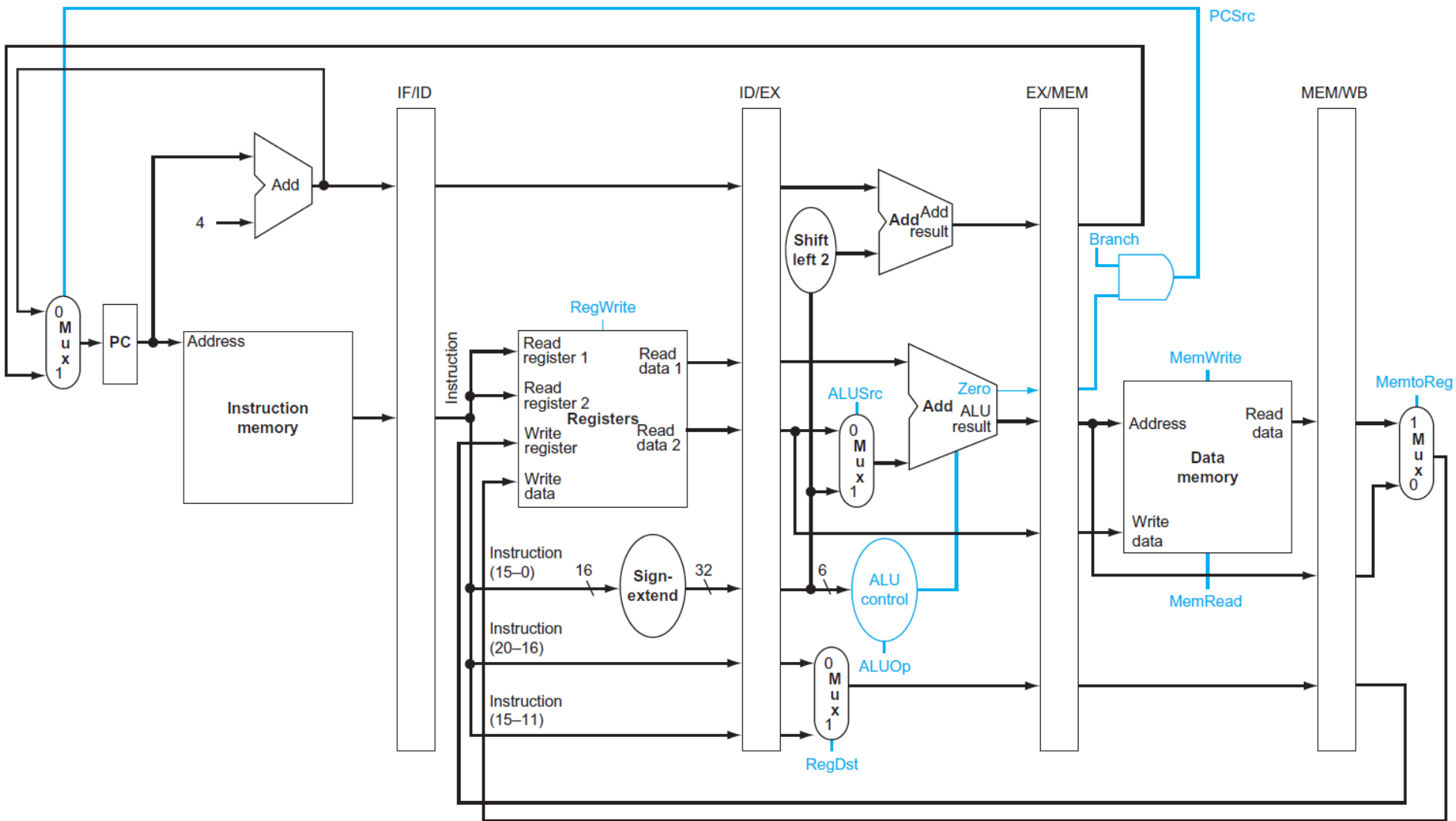


Pipeline diagram

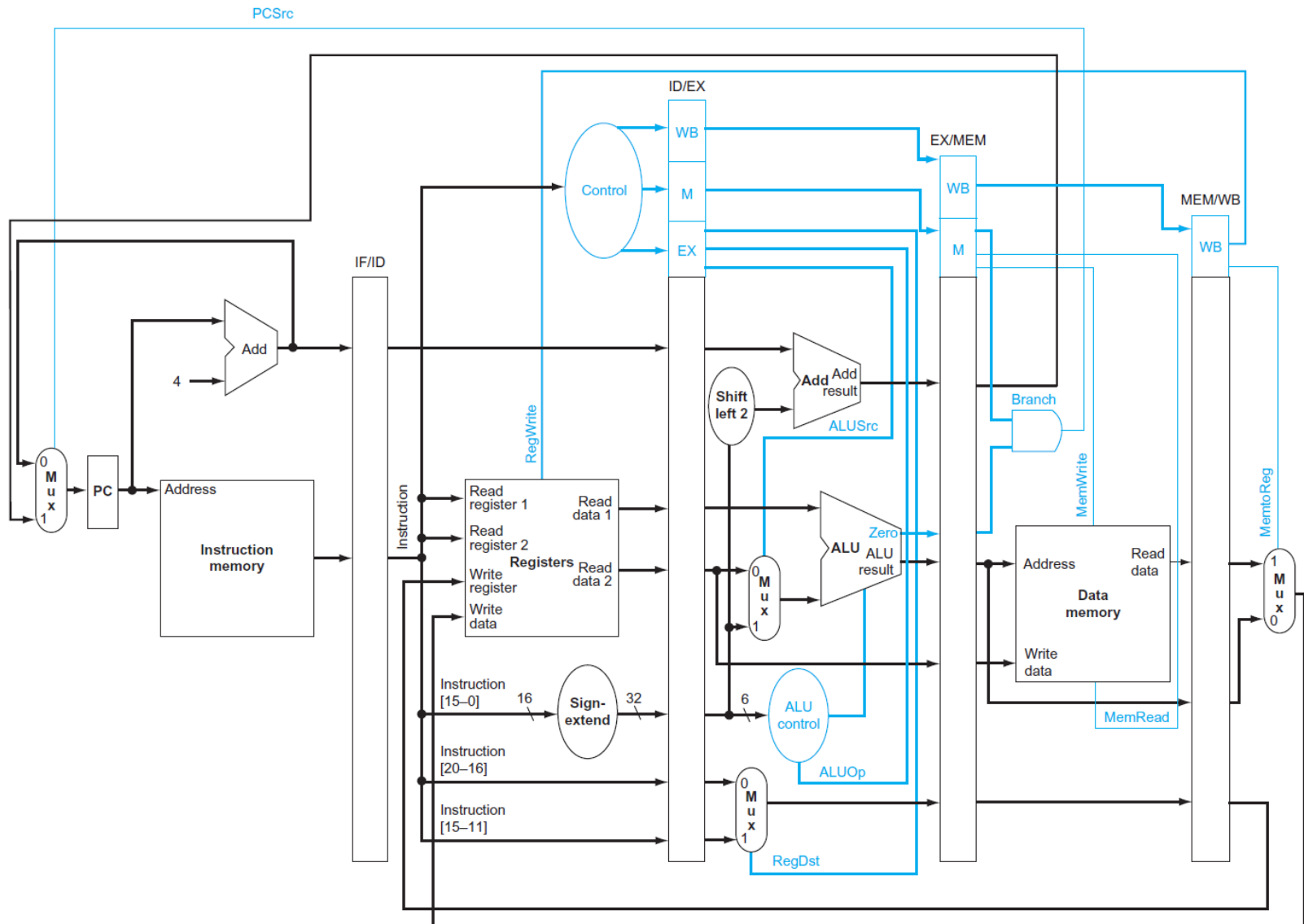
add \$14, \$5, \$6	lw \$13, 24 (\$1)	add \$12, \$3, \$4	sub \$11, \$2, \$3	lw \$10, 20(\$1)
Instruction fetch	Instruction decode	Execution	Memory	Write-back



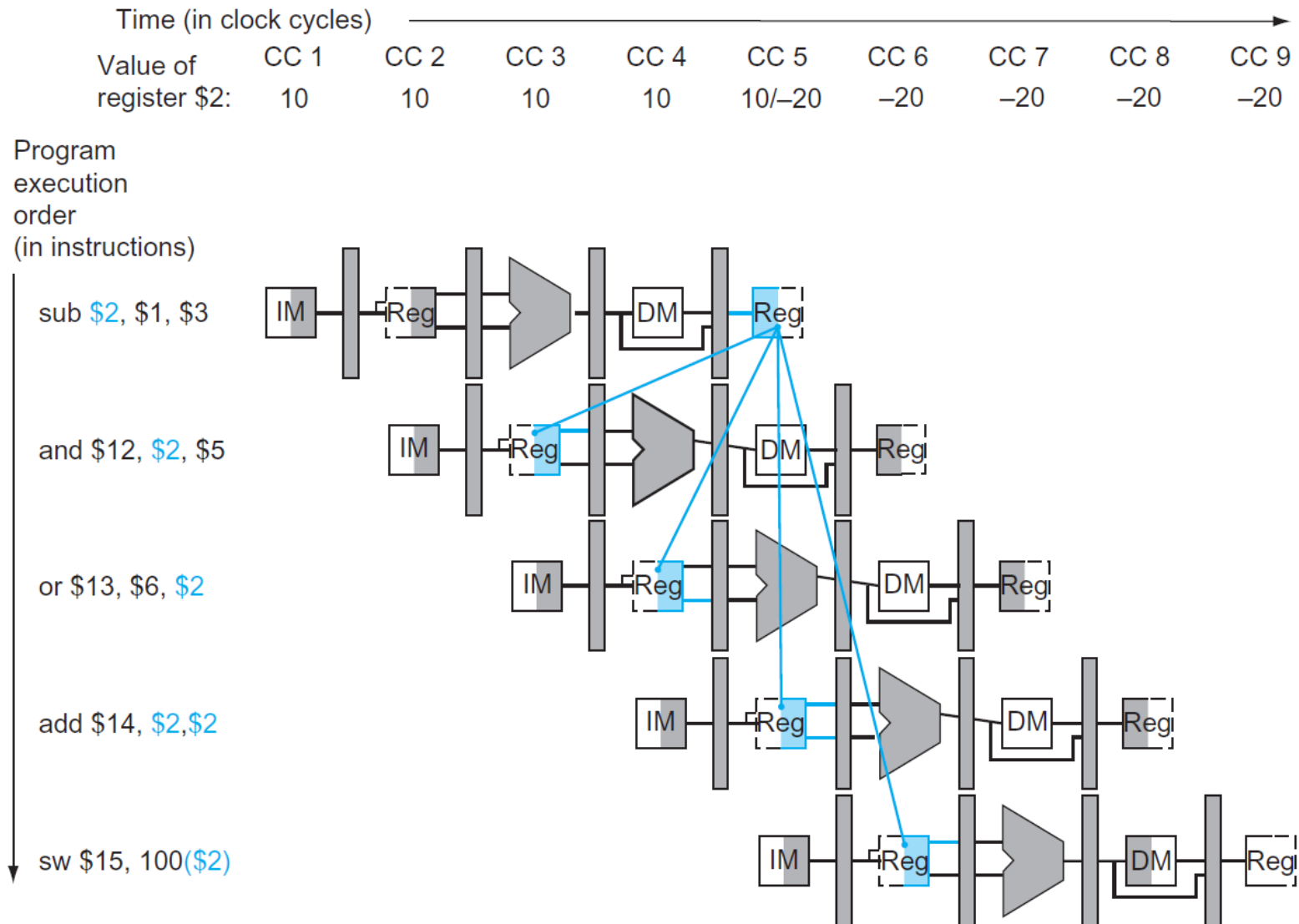
Control signals in pipeline



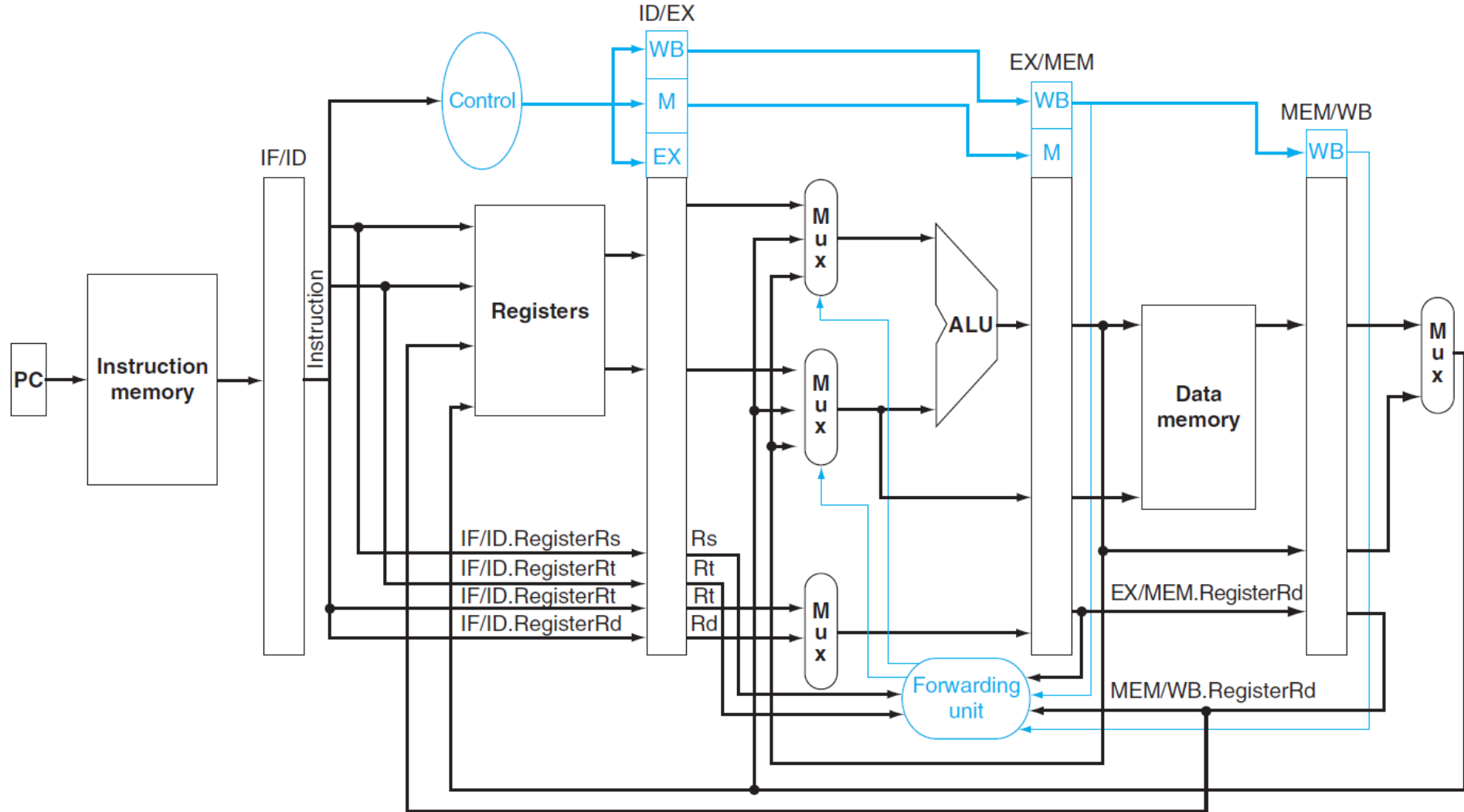
Pipelined control



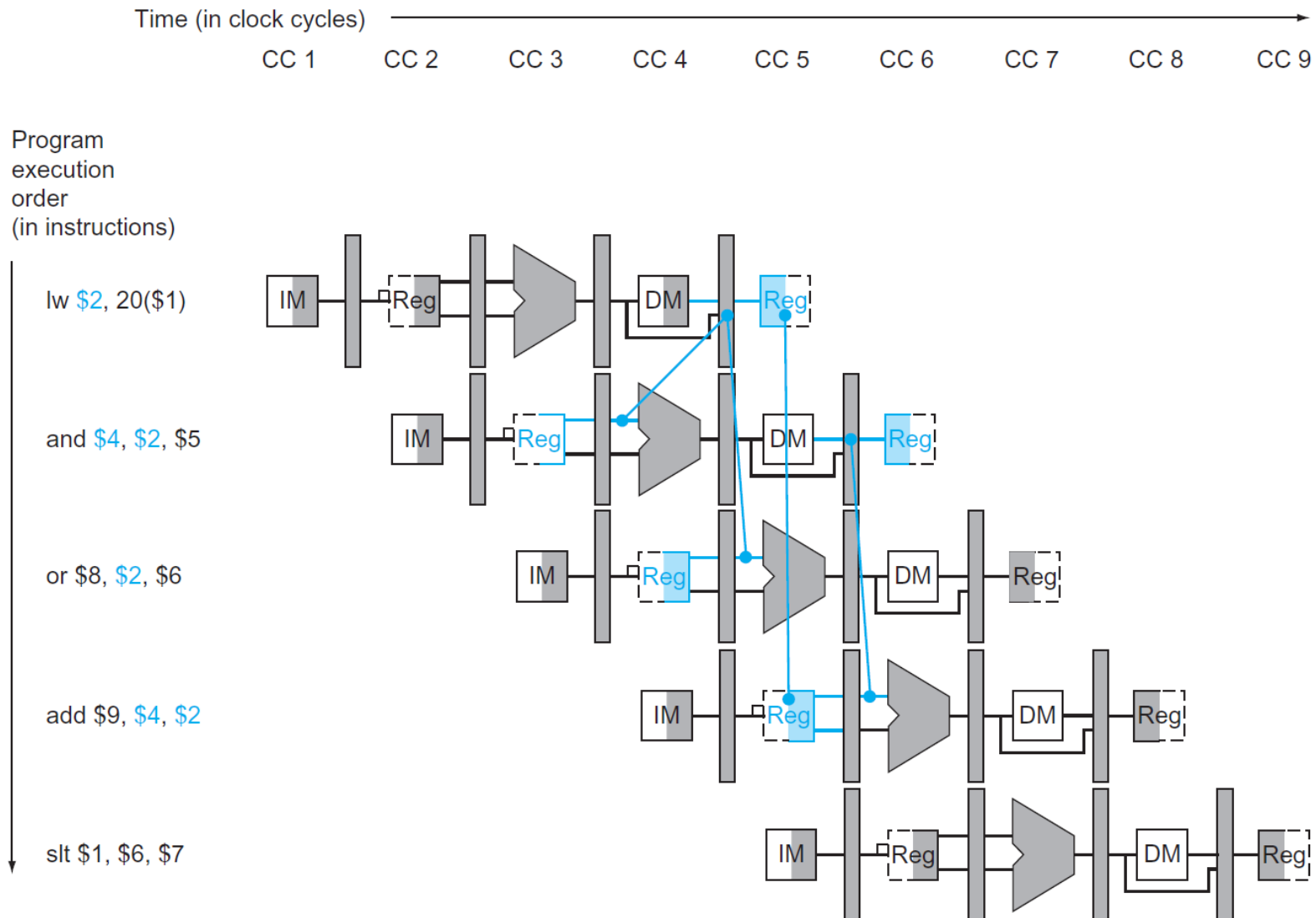
Solving read-after-write hazard



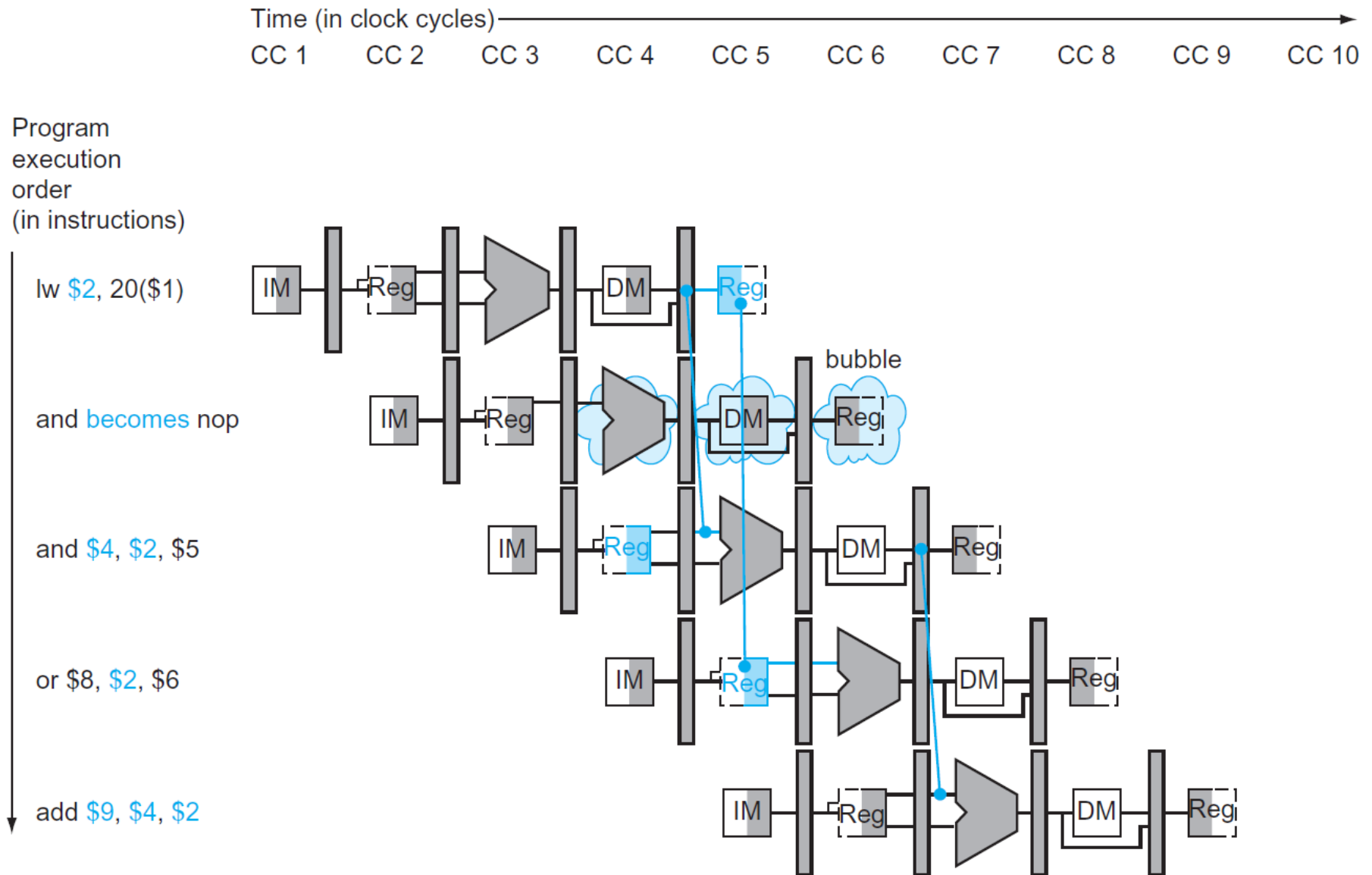
Solving read-after-write hazard



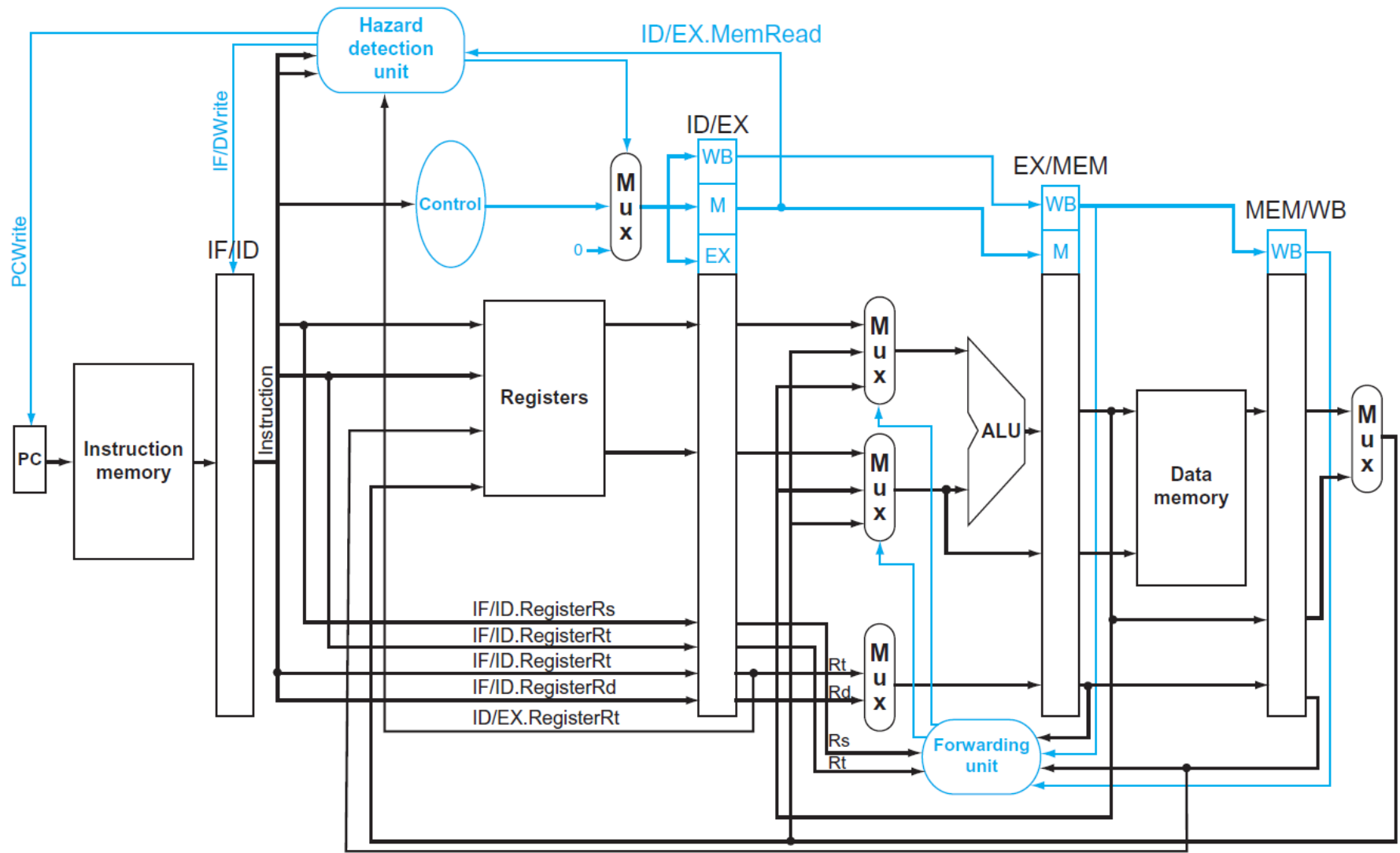
Solving load-use hazard



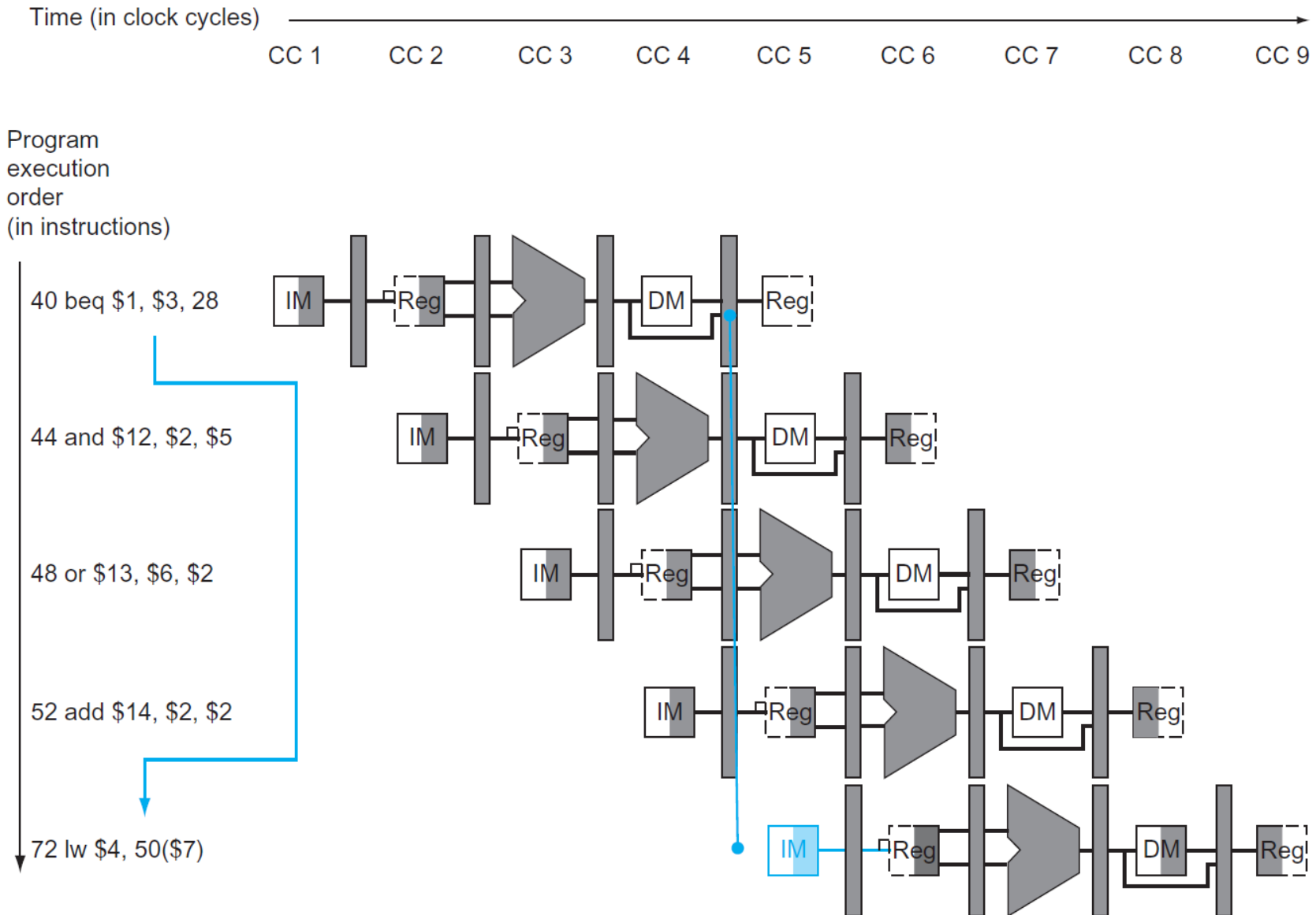
Solving load-use hazard



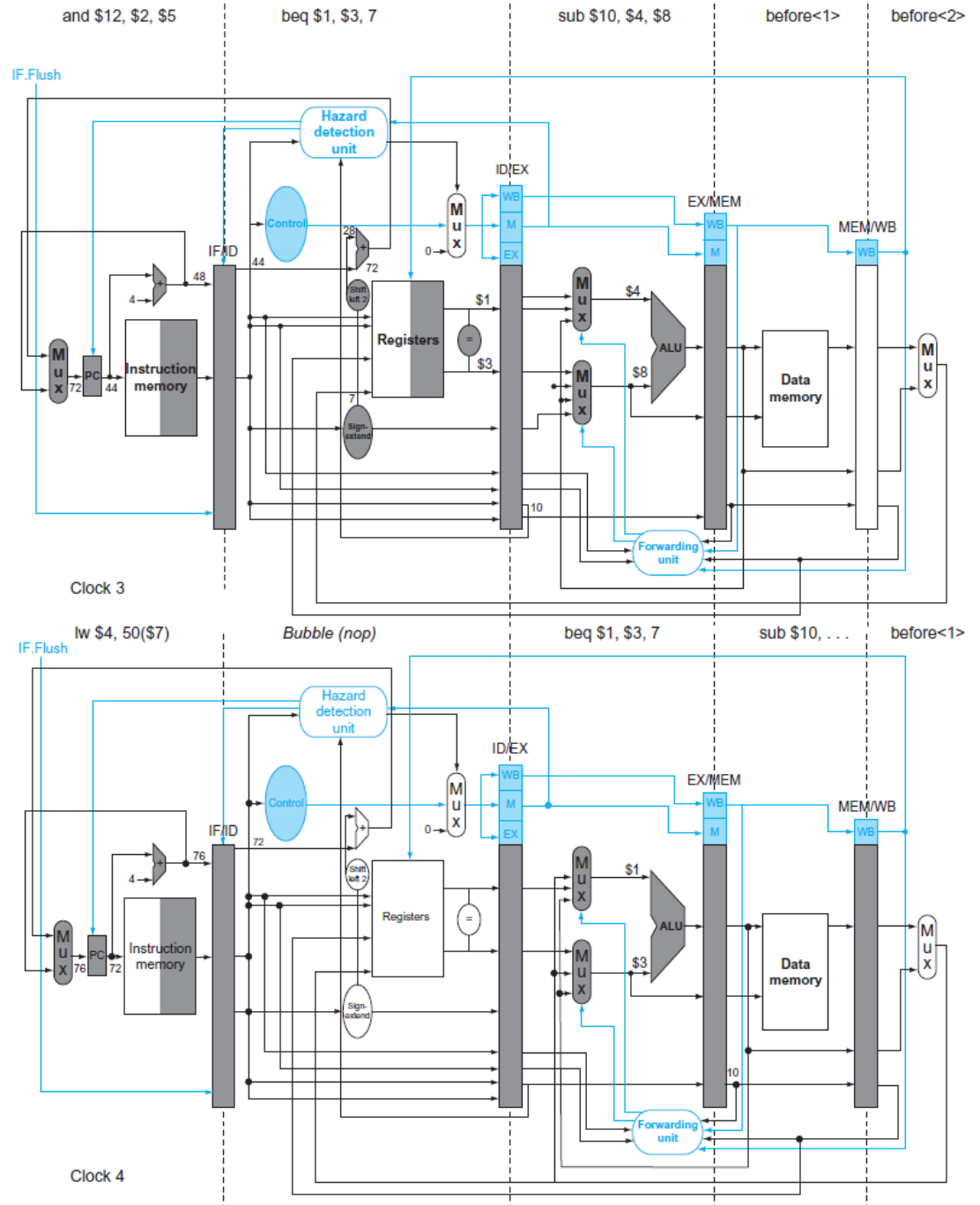
Solving load-use hazard



Solving control hazard



Delayed branching



Summary

- ❑ All modern day processors use pipelining
- ❑ Pipelining doesn't help **latency** of single task, it helps **throughput** of entire workload
- ❑ Potential speedup: a CPI of 1 and a fast CC
- ❑ Must detect and resolve hazards
 - ❑ Stalling negatively affects CPI (makes CPI less than the ideal of 1)

Example

- ❑ Detect stall in the code snippets below

```
lw    $t0, 0($t0)           add    $t1, $t0, $t0
add   $t1, $t0, $t0         addi   $t2, $t0, #5
                                addi   $t4, $t1, #5

                                addi   $t1, $t0, #1
                                addi   $t2, $t0, #2
                                addi   $t3, $t0, #2
                                addi   $t3, $t0, #4
                                addi   $t5, $t0, #5
```